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Daggett et al.

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(54) **TRUNNION-TO-DISK CONNECTION FOR
AN OPEN FAN CONFIGURATION
AIRCRAFT POWERPLANT**

(58) **Field of Classification Search**

CPC F02K 3/06; F04D 29/34; F04D 29/601

See application file for complete search history.

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Related U.S. Application Data

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17, 2023, now Pat. No. 11,913,408.

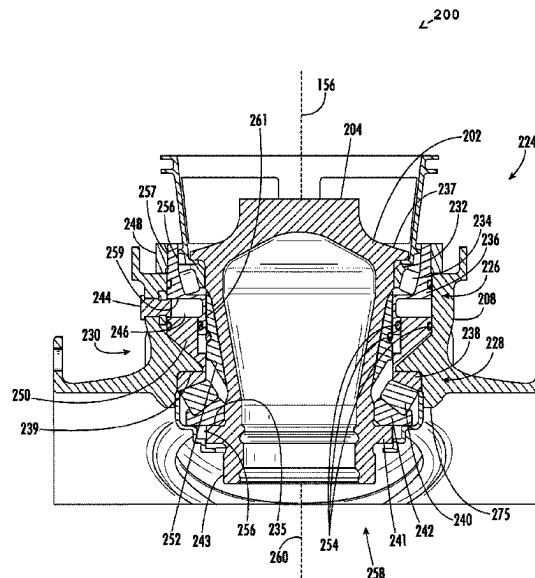
(51) **Int. Cl.**
F02K 3/06 (2006.01)
F04D 29/34 (2006.01)
F04D 29/60 (2006.01)

(52) **U.S. Cl.**
CPC **F02K 3/06** (2013.01); **F04D 29/34**
(2013.01); **F04D 29/601** (2013.01)

(57) **ABSTRACT**

A trunnion-to-disk connection for use on an open fan configuration of a gas turbine engine may include an integral trunnion and blade spar inserted through a trunnion aperture of a fan disk and supported by top bearing and a bottom bearing. A cavity can be provided between a trunnion of the integral trunnion and blade spar and the fan disk, as well as between the top bearing and bottom bearing. Pressurized hydraulic fluid can be supplied to the cavity to urge the integral trunnion and blade spar in a direction to preload the bearings. Prior to pressurization, and prior to installation of the bottom bearing, the trunnion can be inserted into a trunnion aperture of the fan disk such that an end of the trunnion extends past the fan disk to provide sufficient space to insert the bottom bearing from within the open interior of the fan disk.

17 Claims, 26 Drawing Sheets



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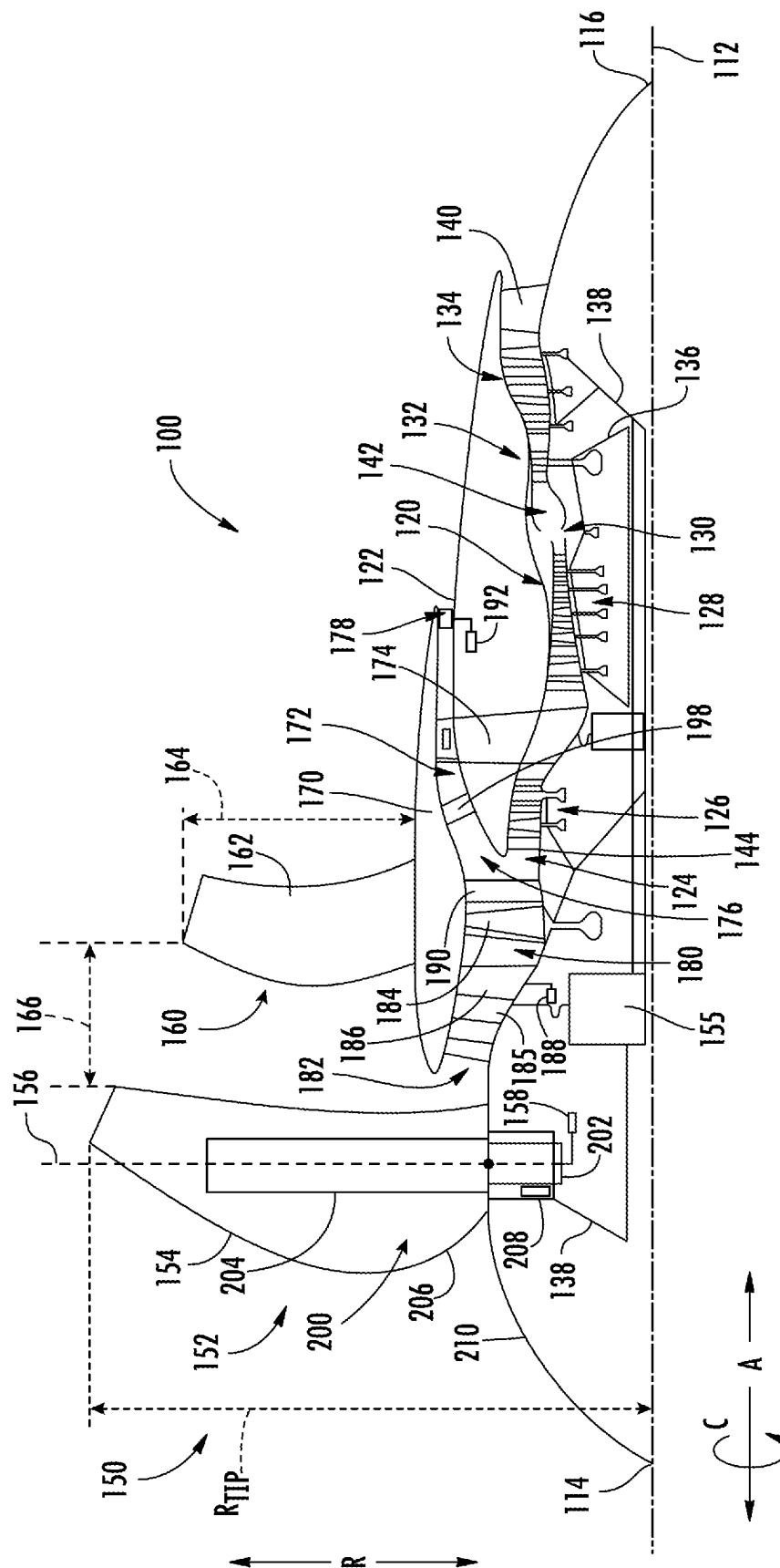
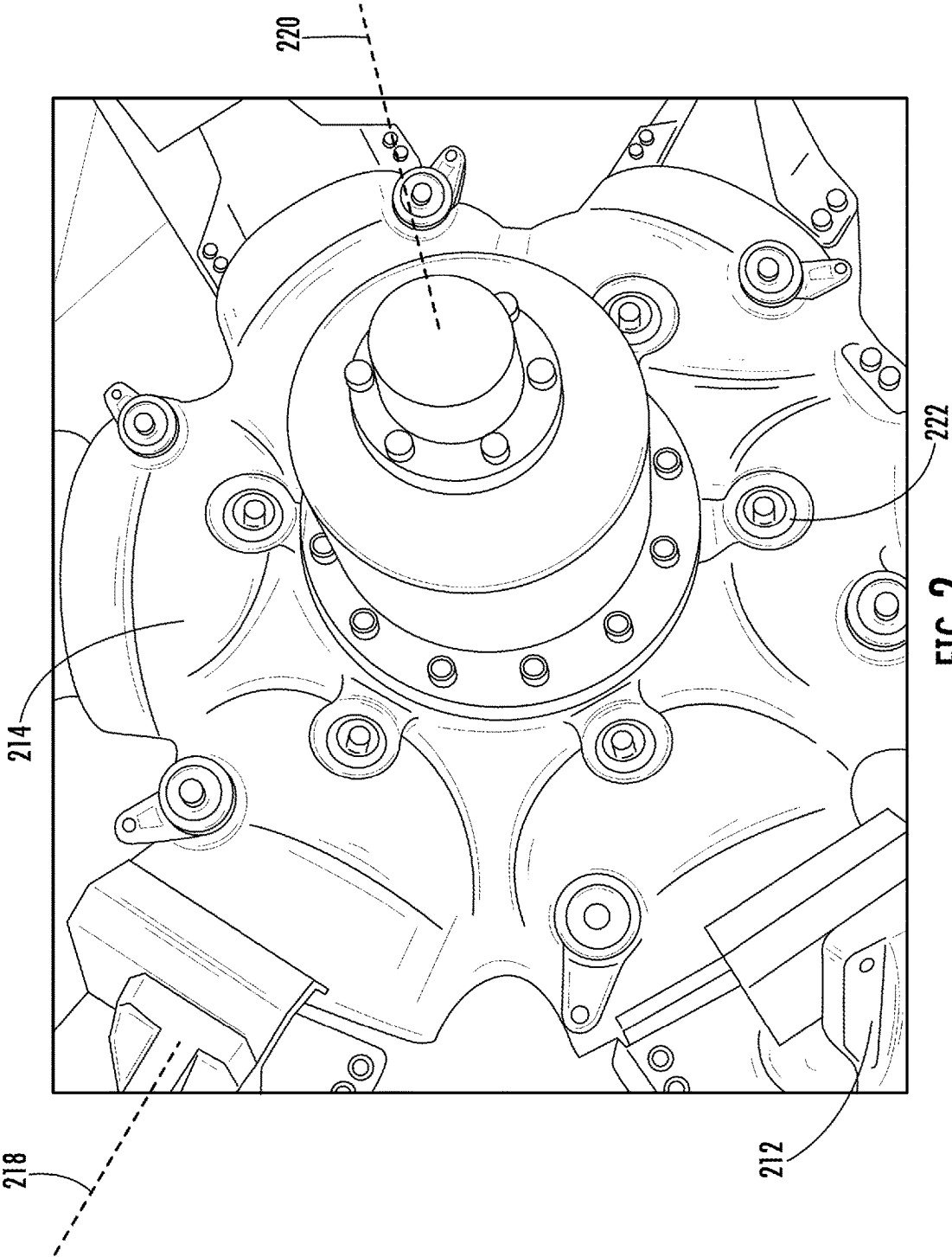


FIG. 1



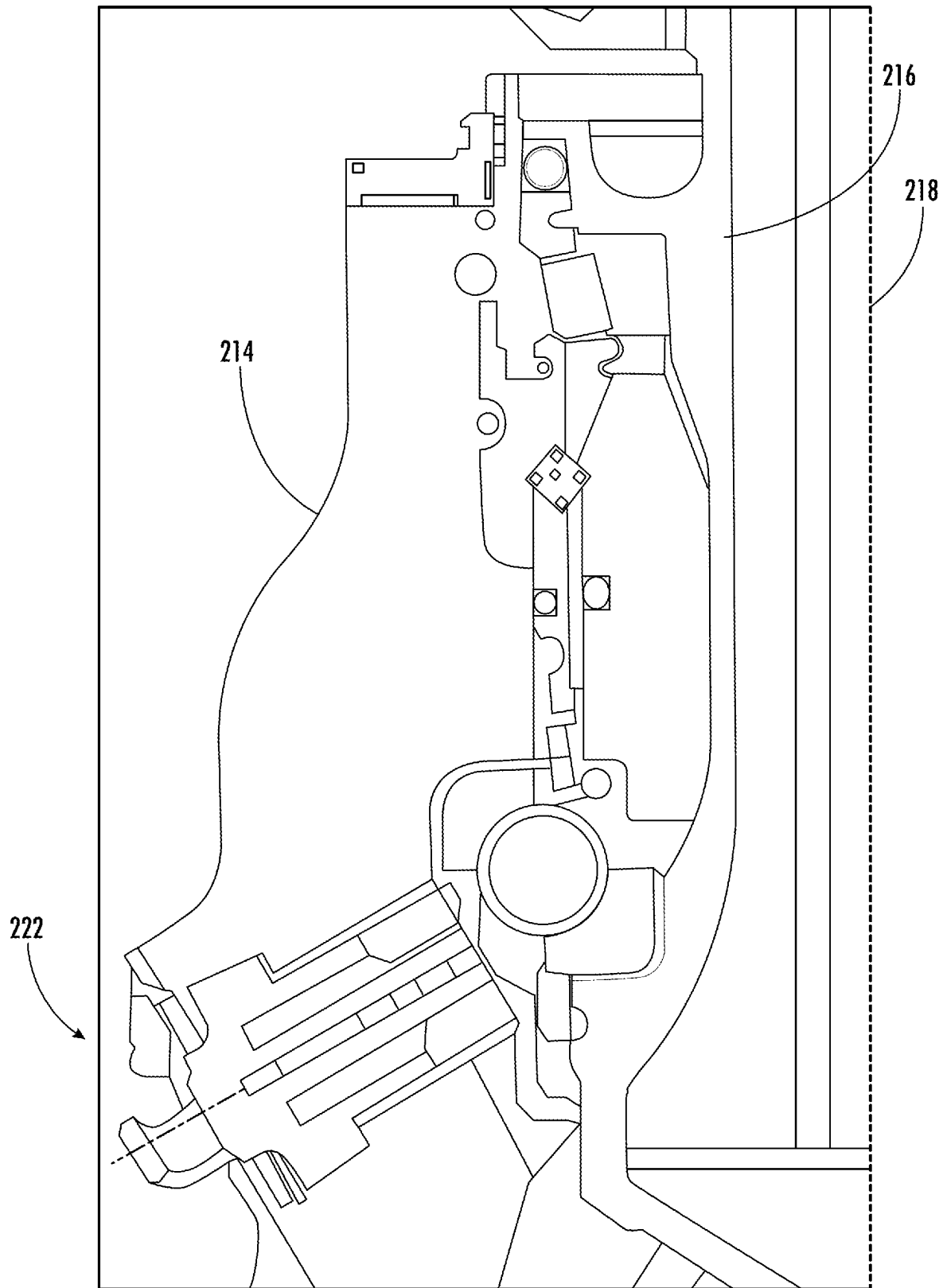


FIG. 3
PRIOR ART

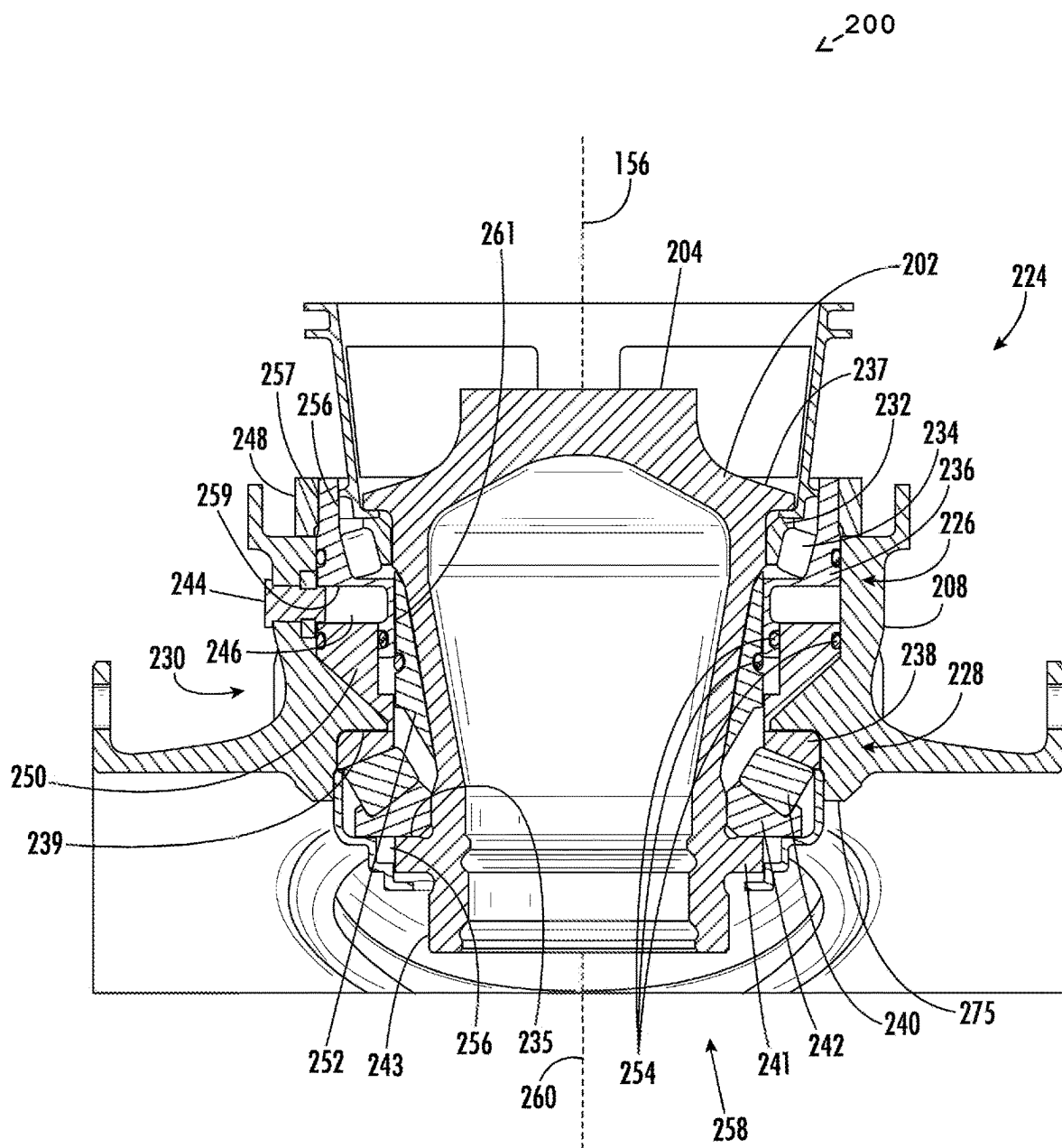


FIG. 4

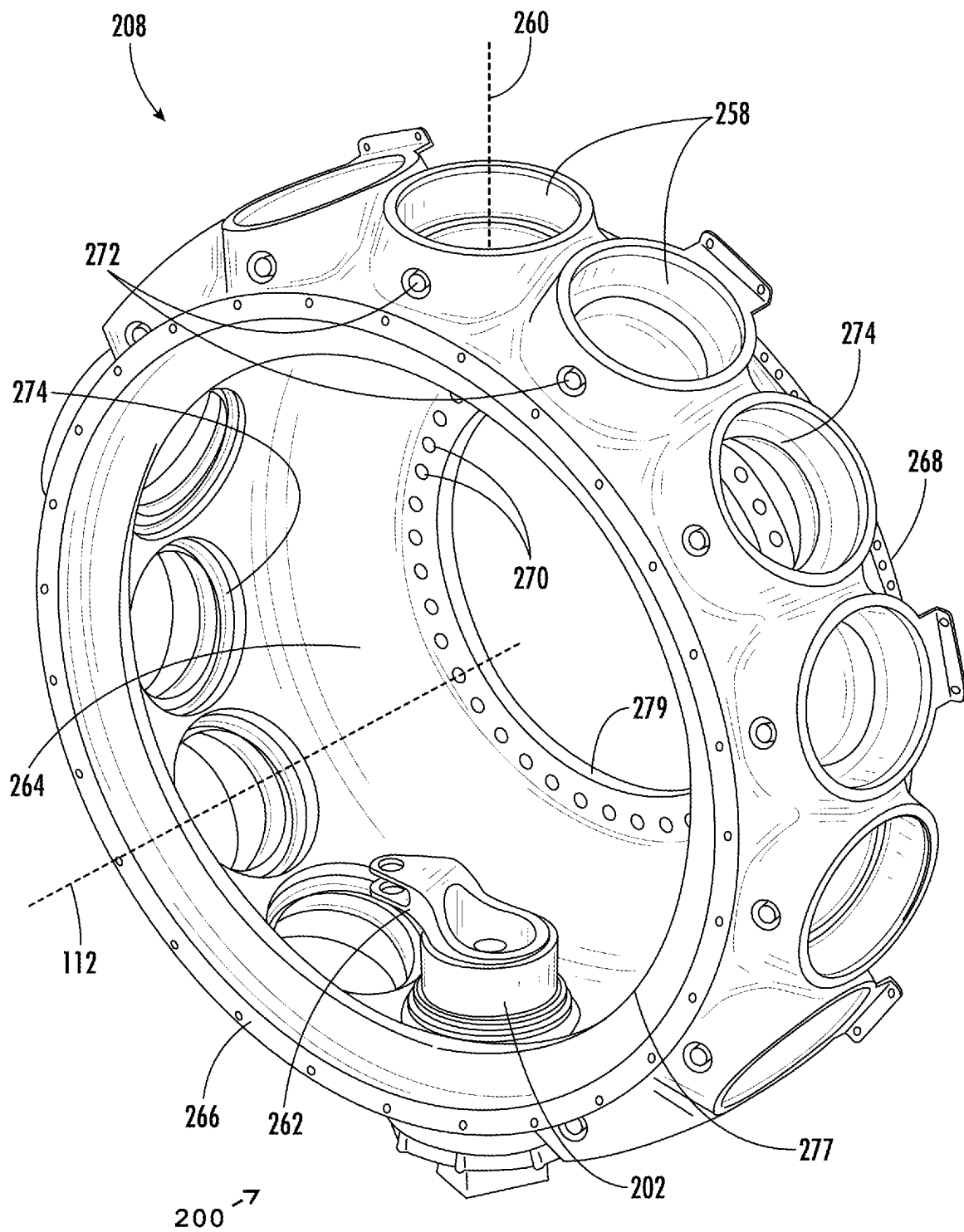


FIG. 5

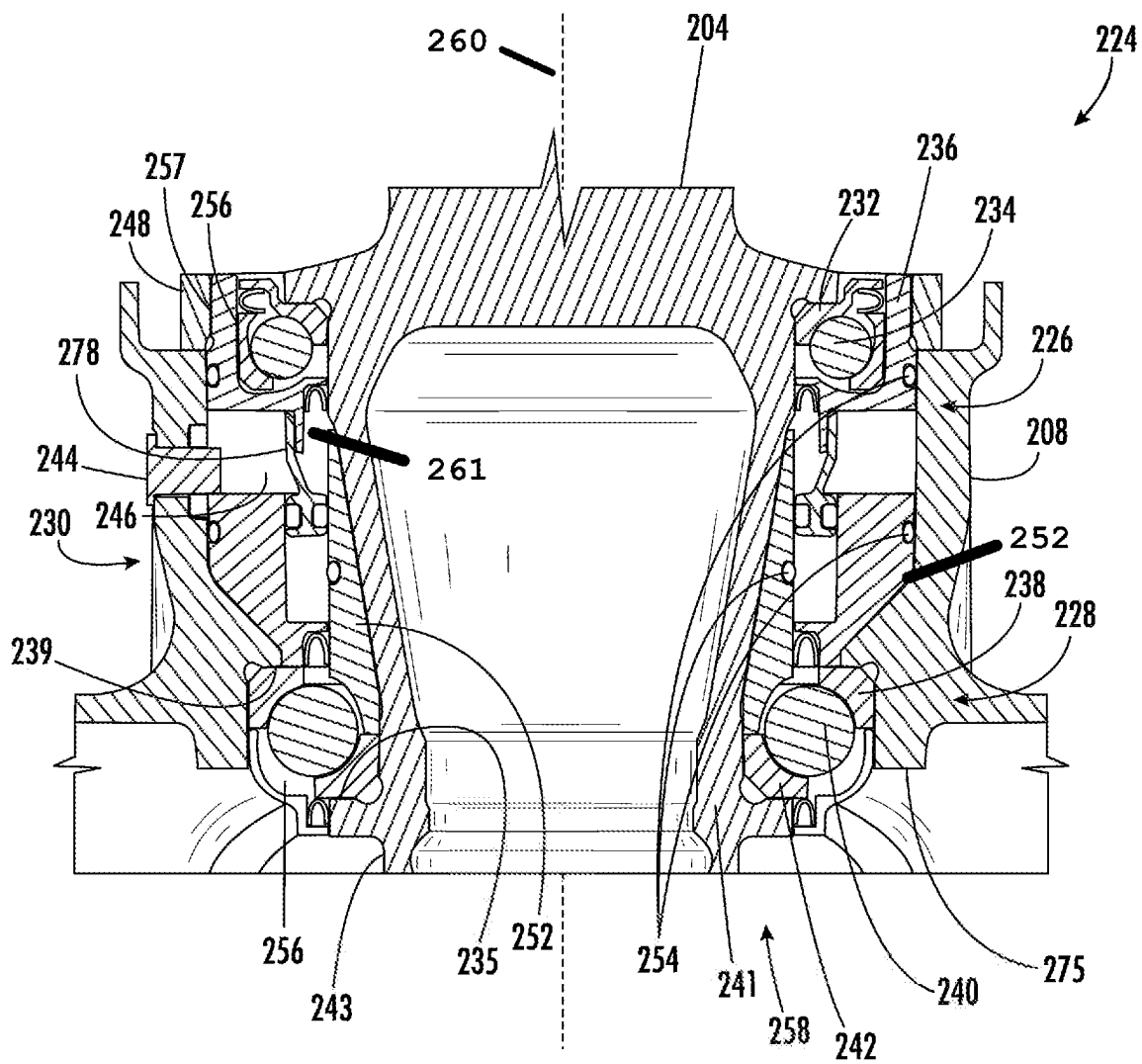
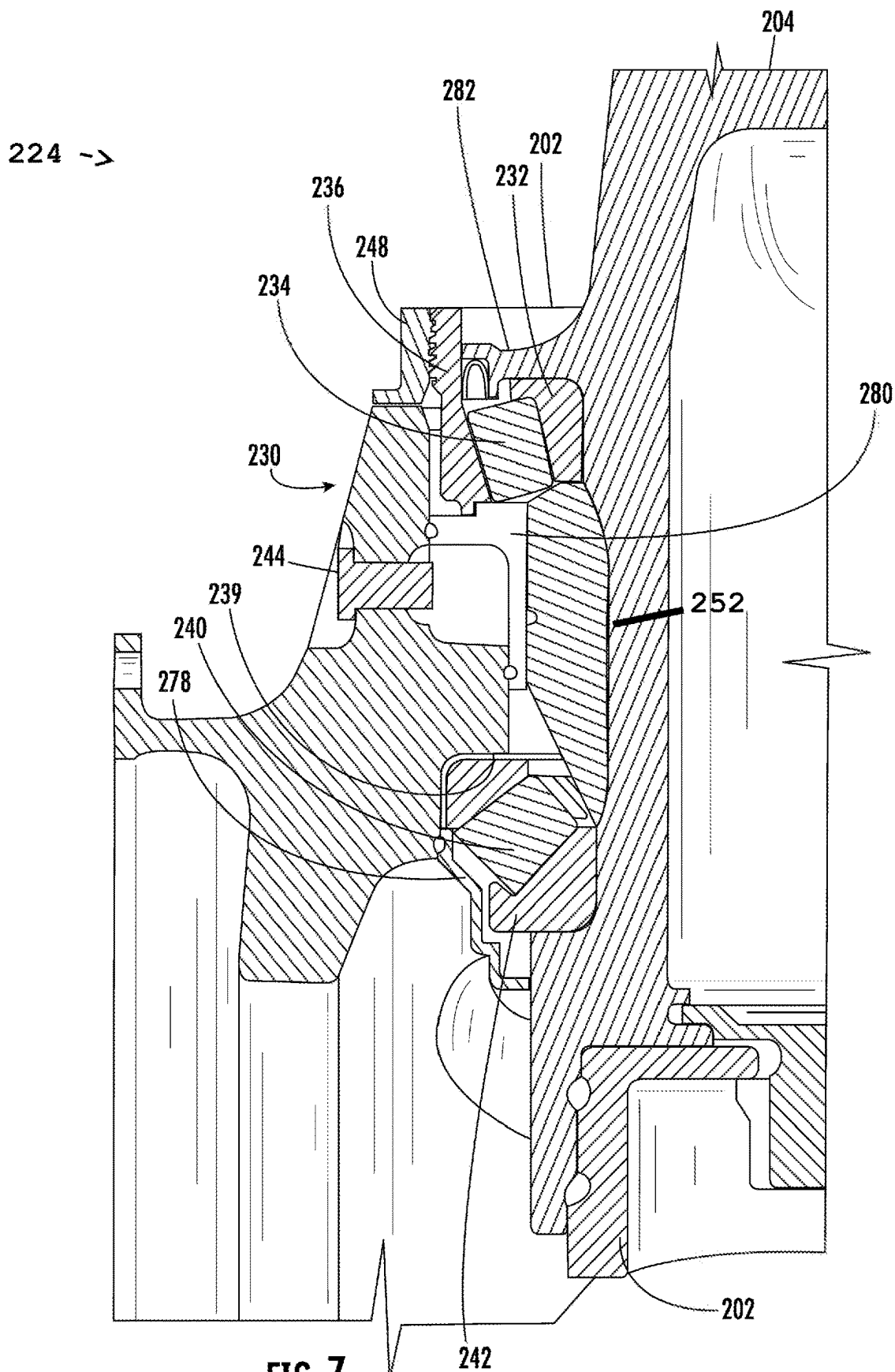
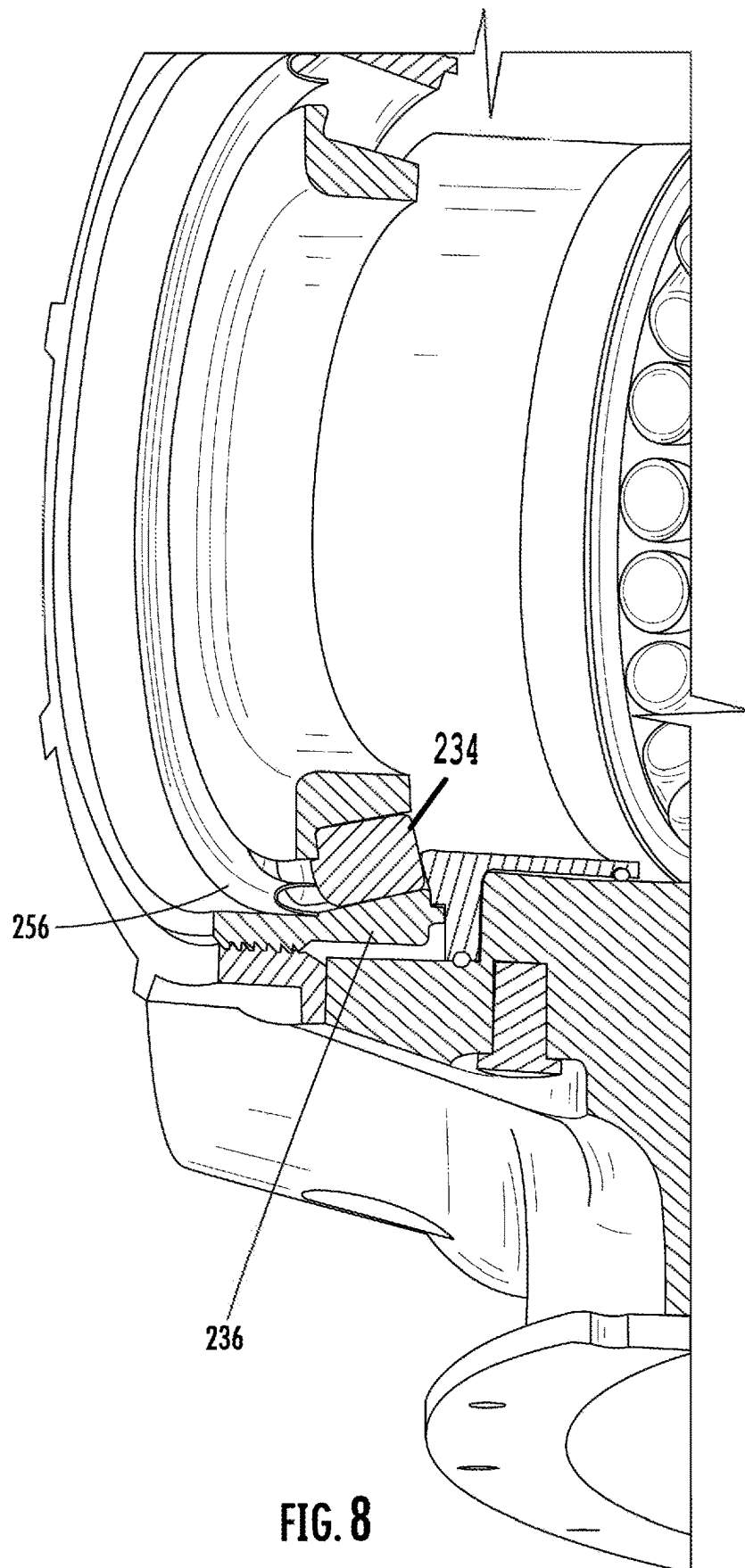


FIG. 6





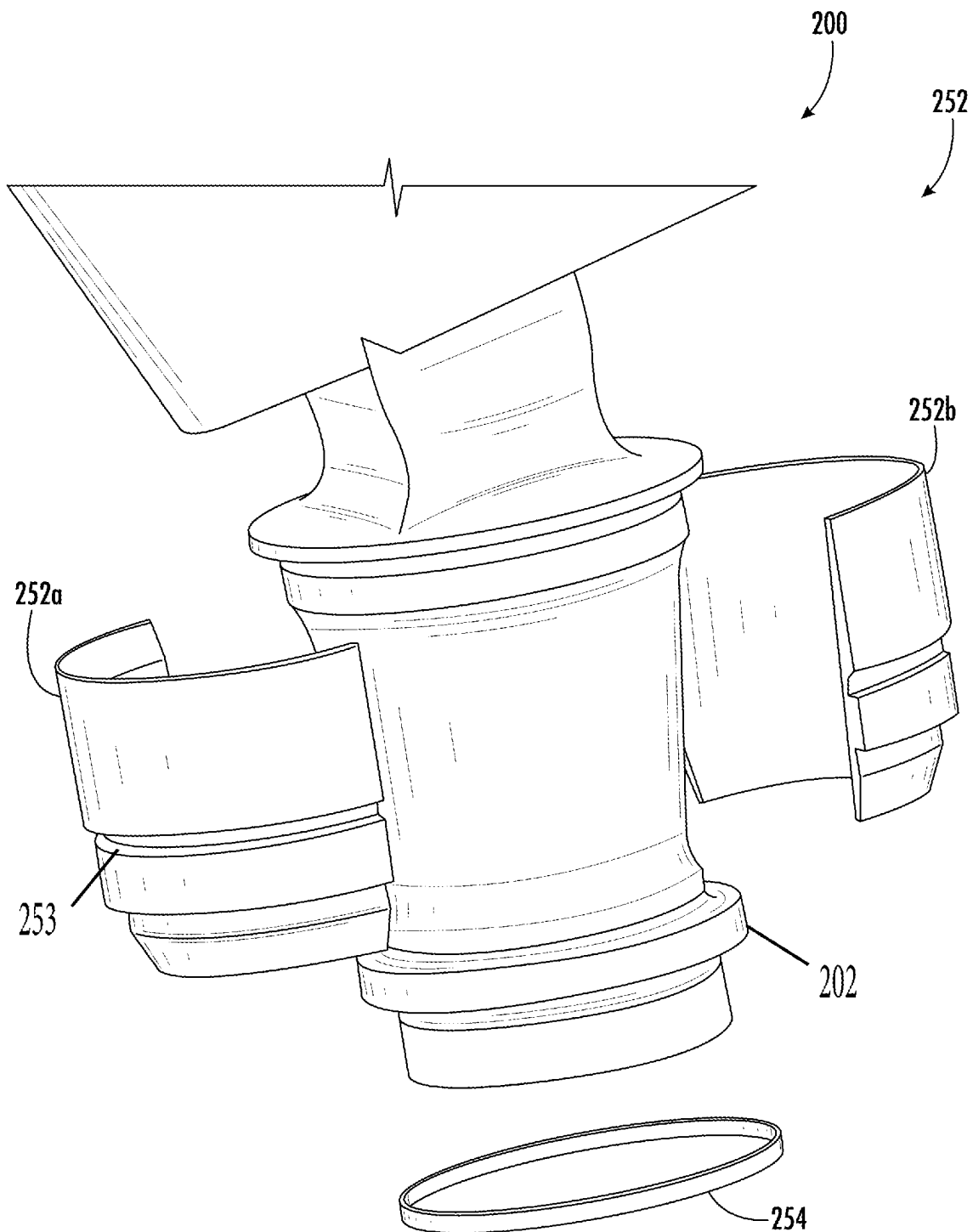


FIG. 9

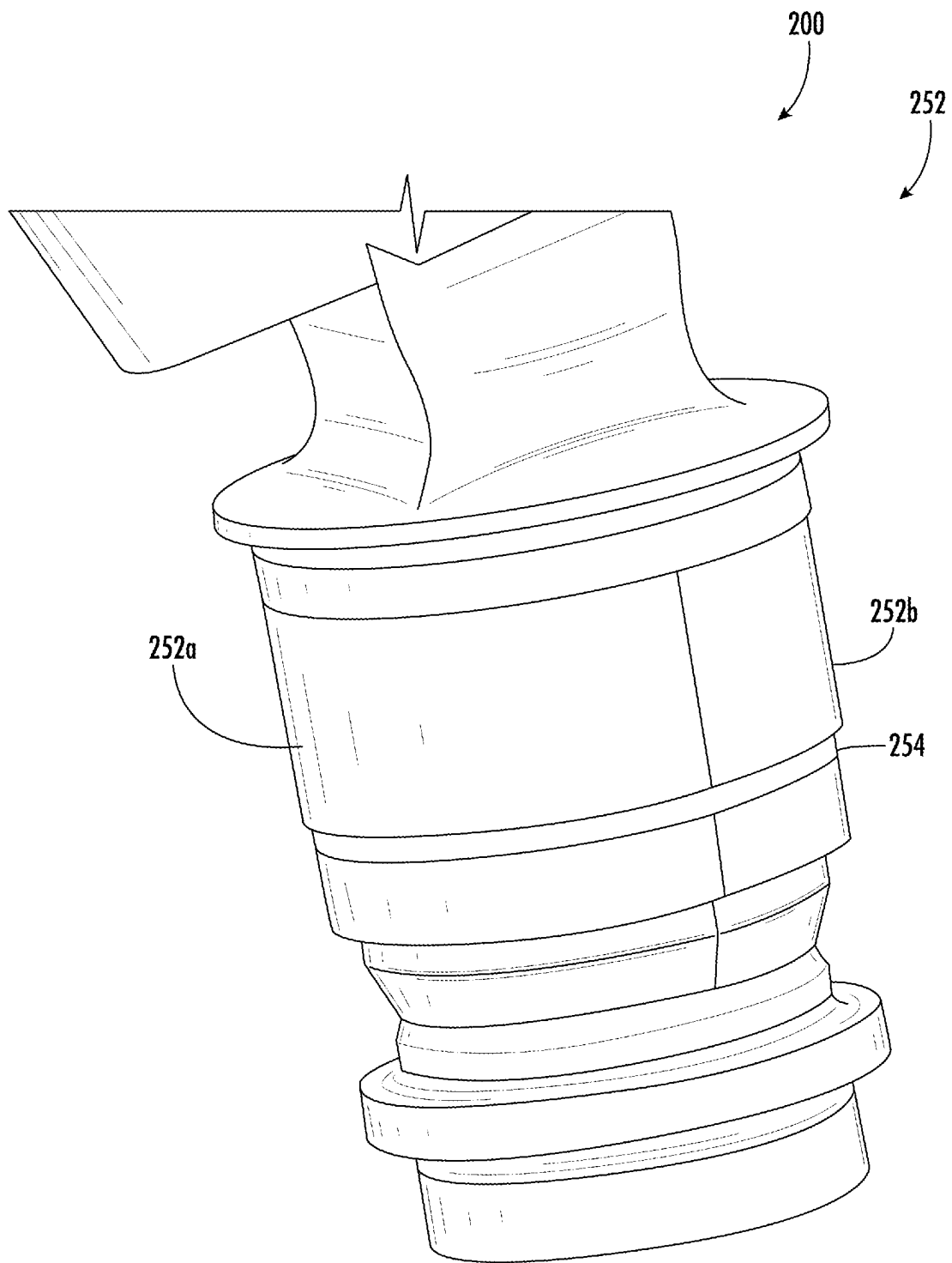


FIG. 10

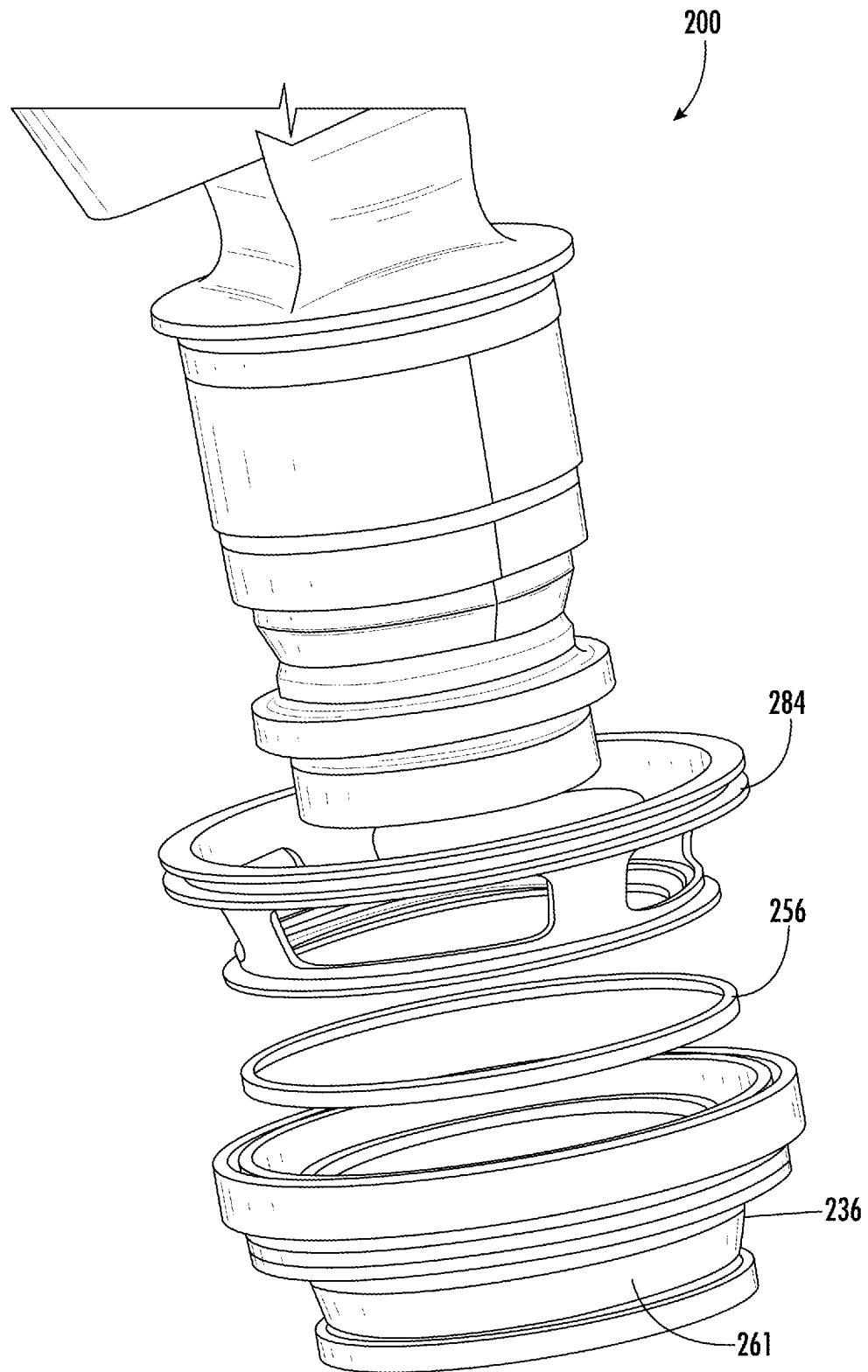


FIG. 11

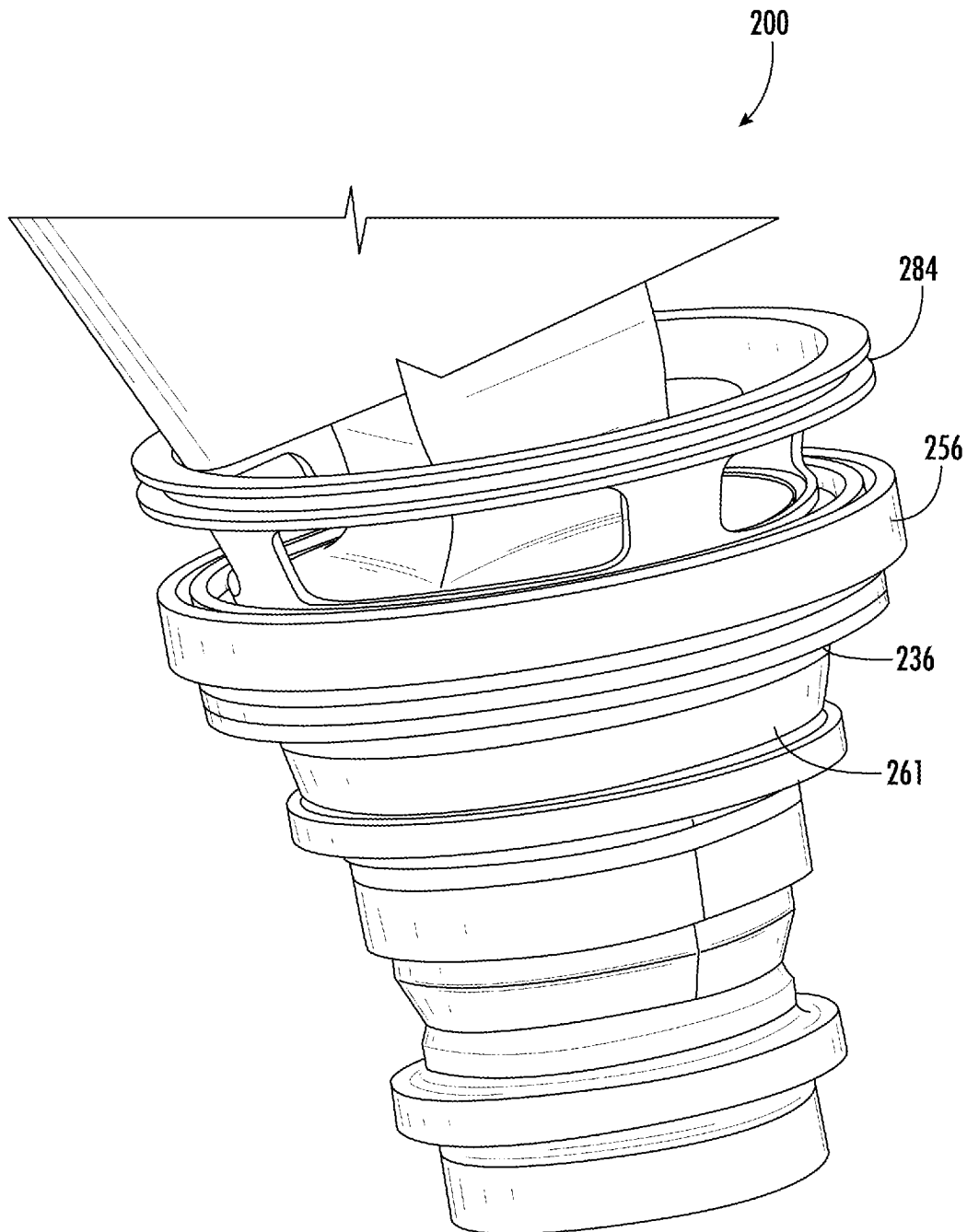


FIG. 12

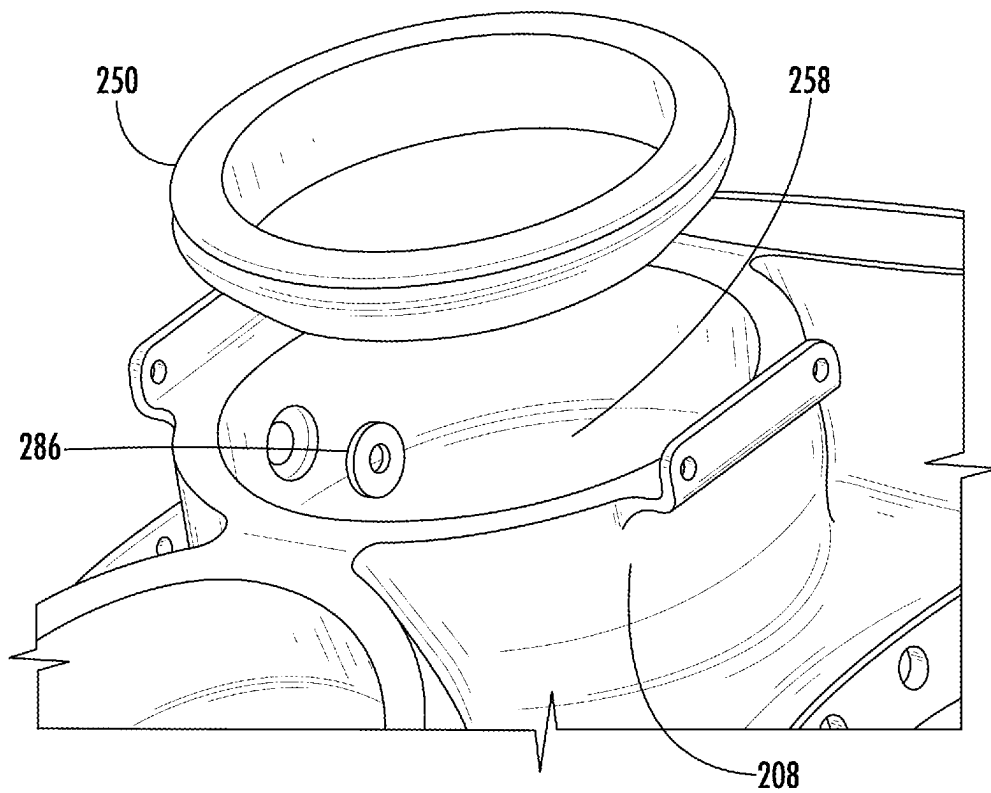


FIG. 13

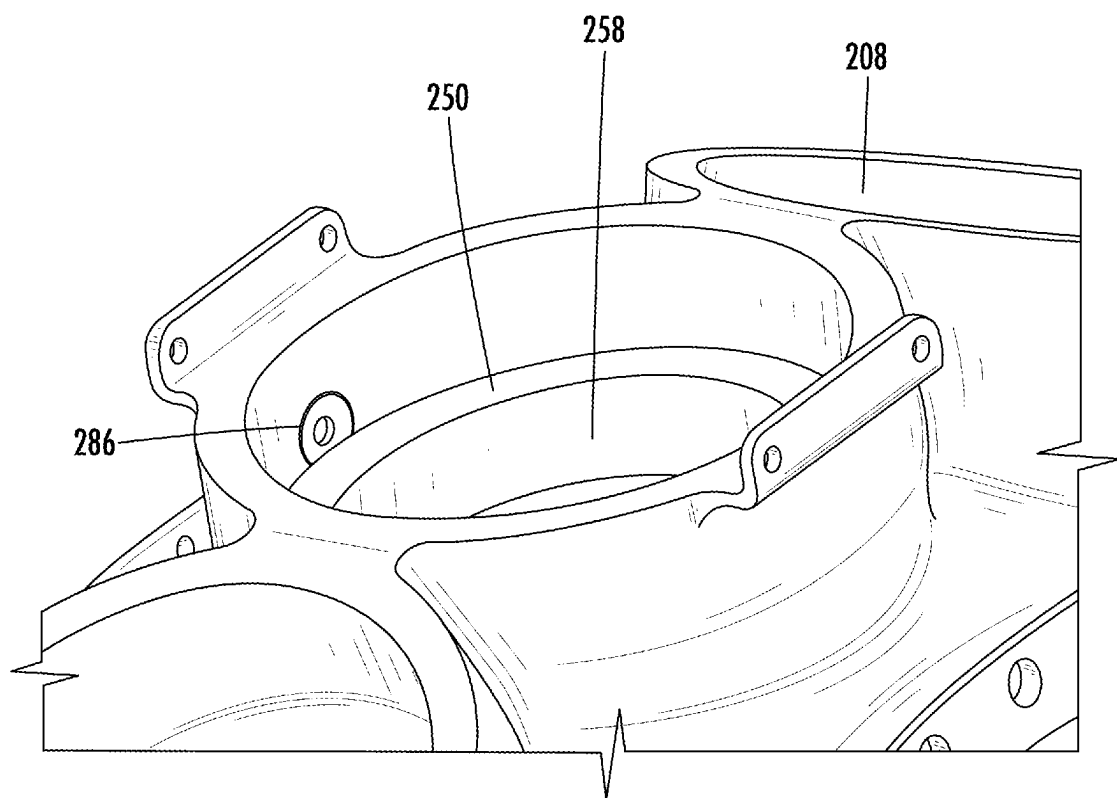
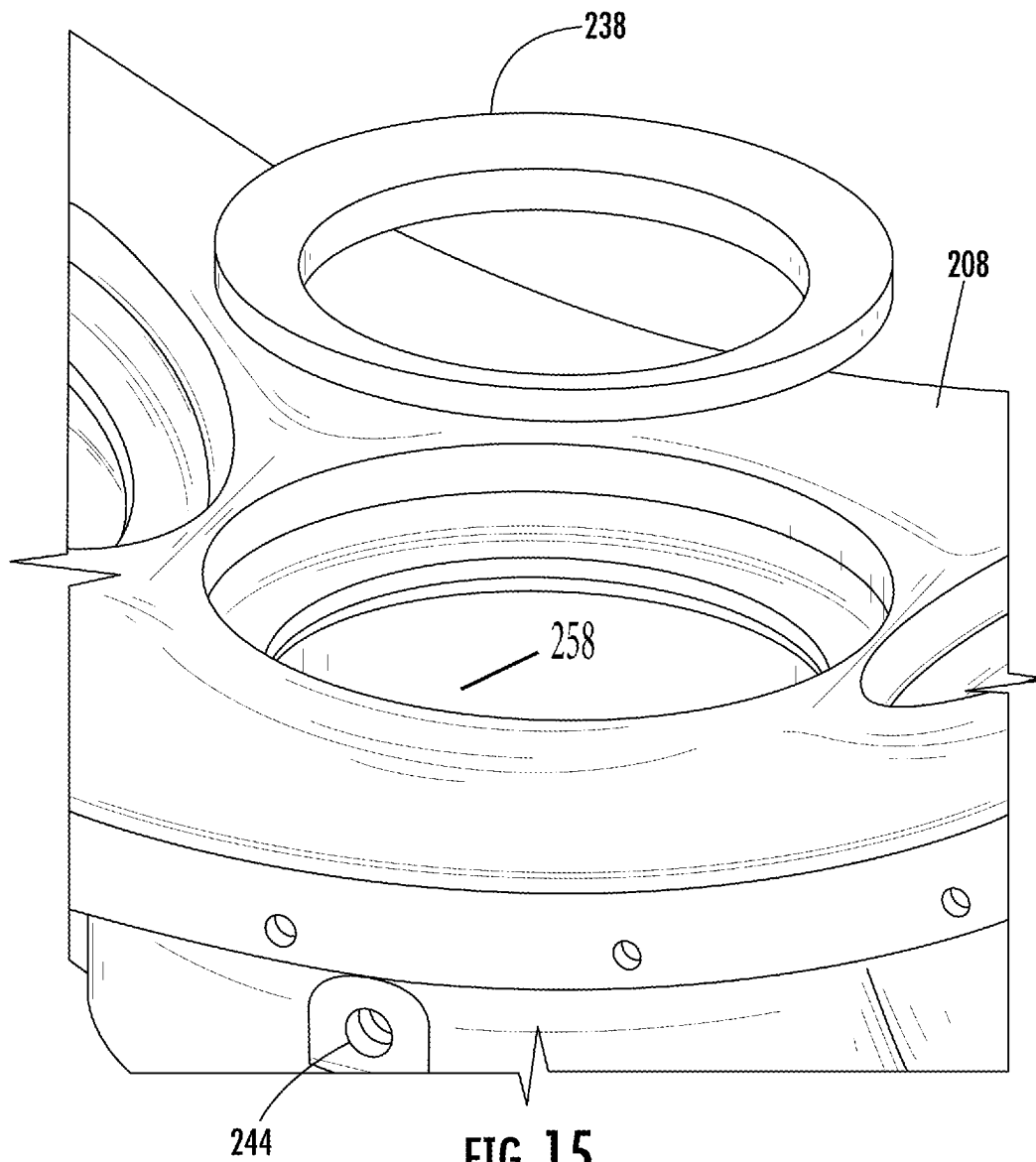


FIG. 14



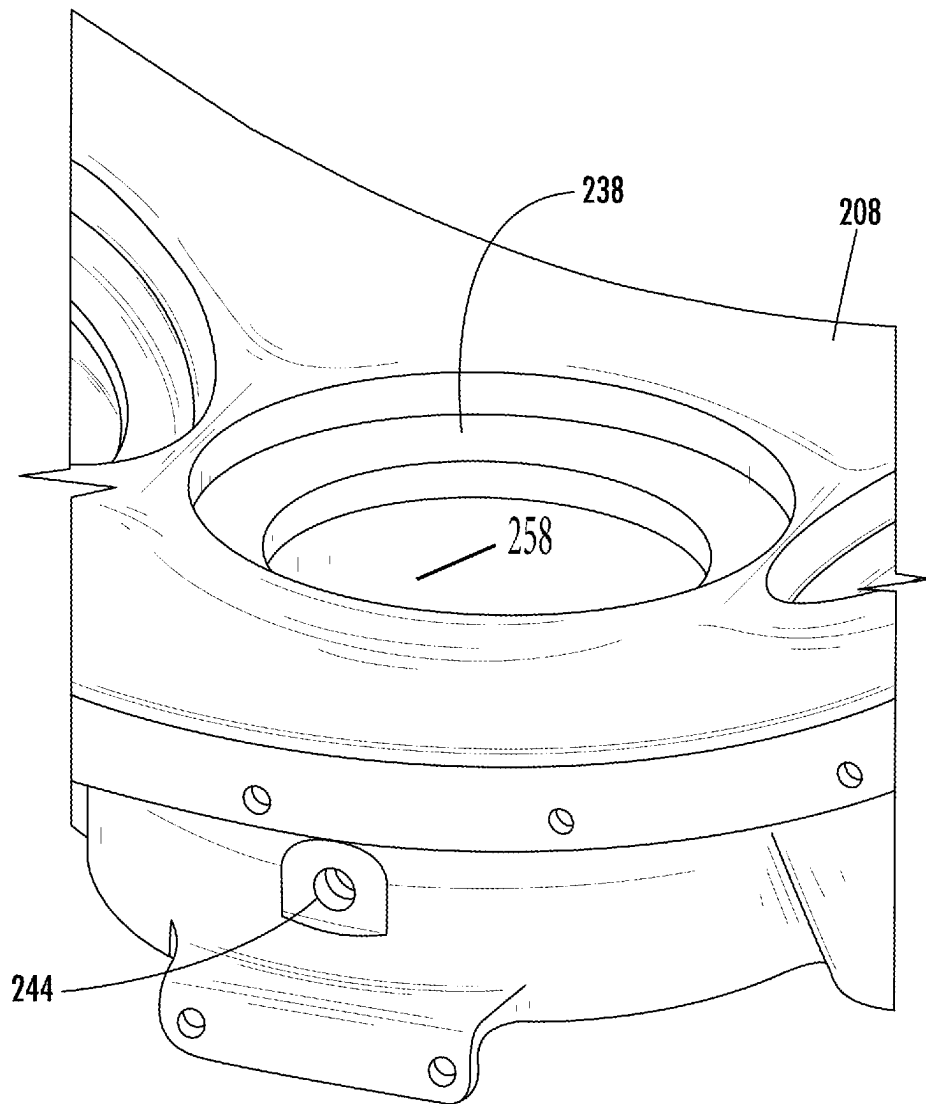


FIG. 16

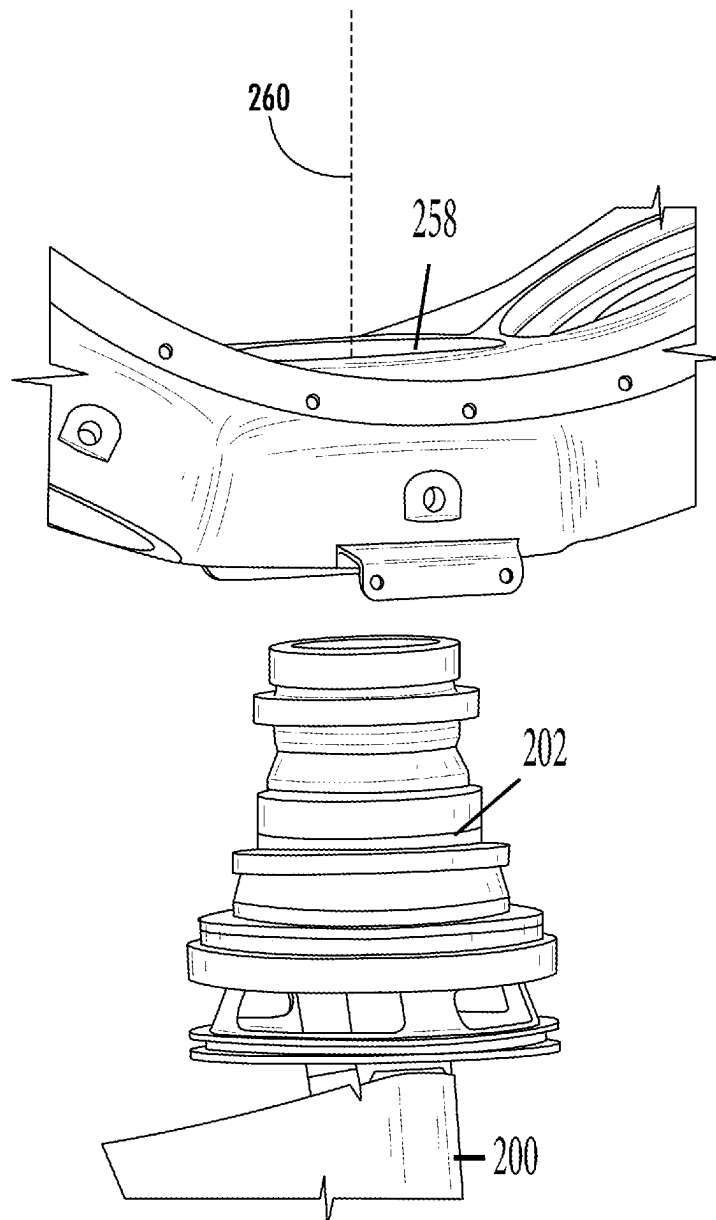


FIG. 17

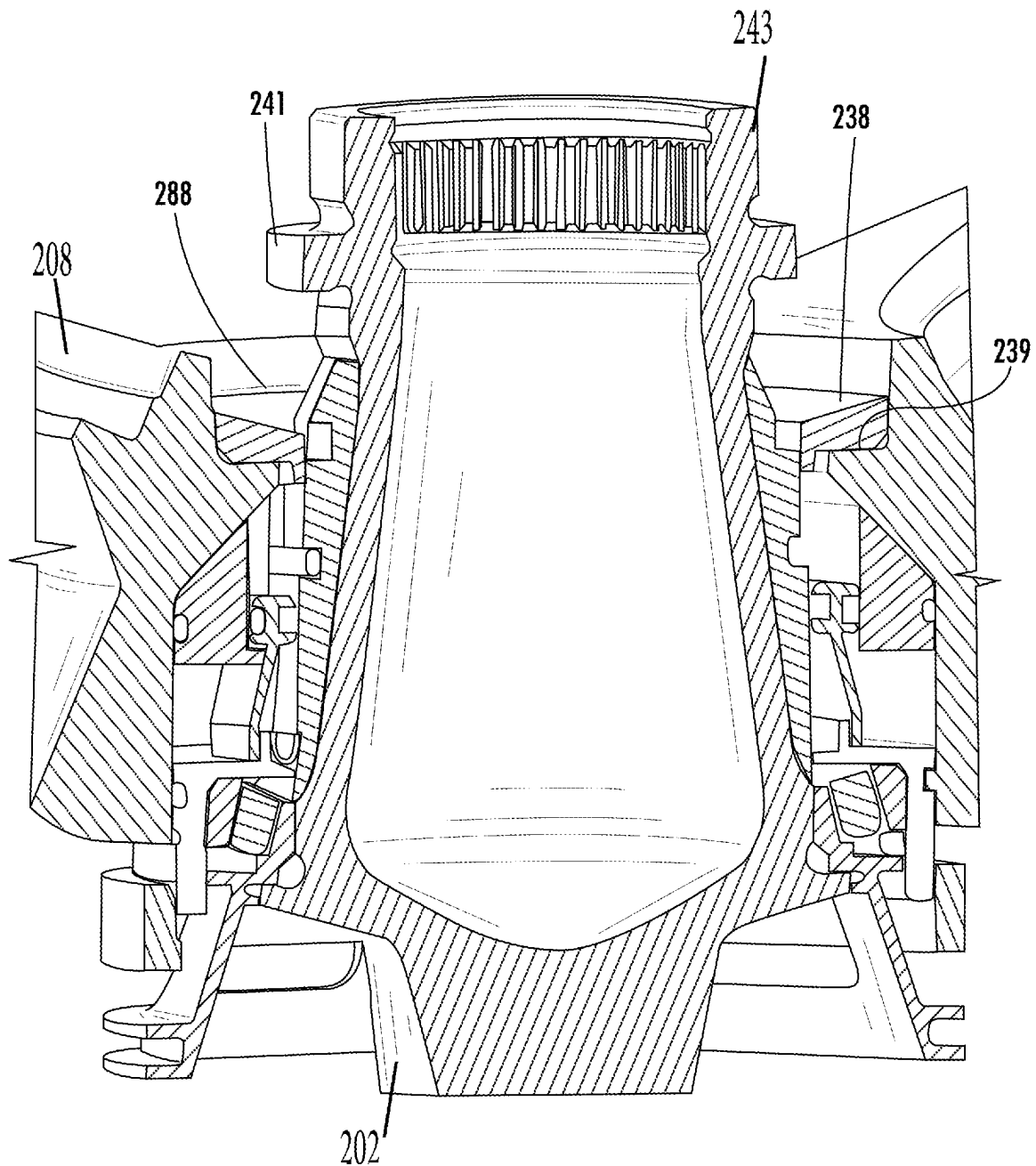


FIG. 18

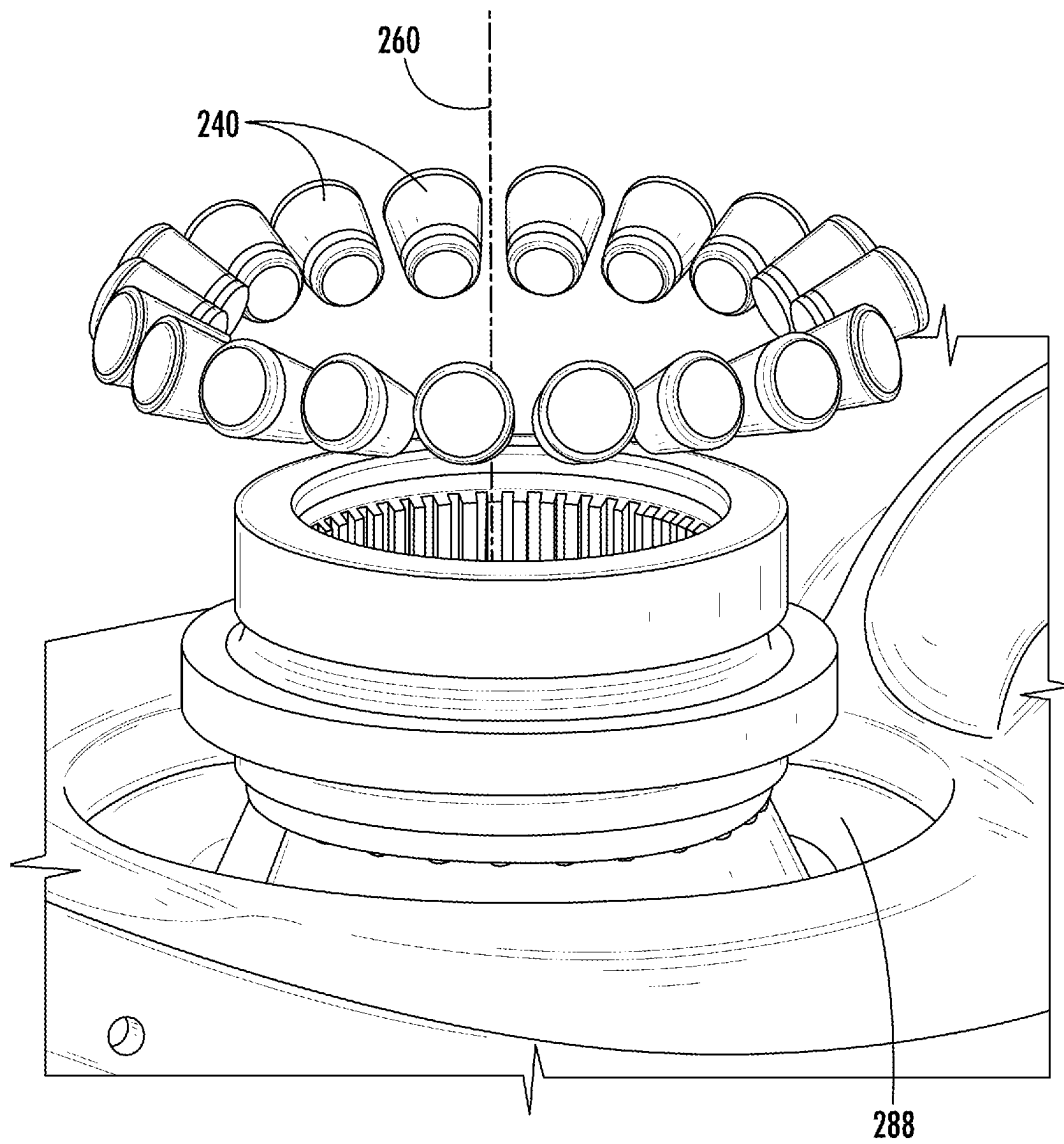


FIG. 19

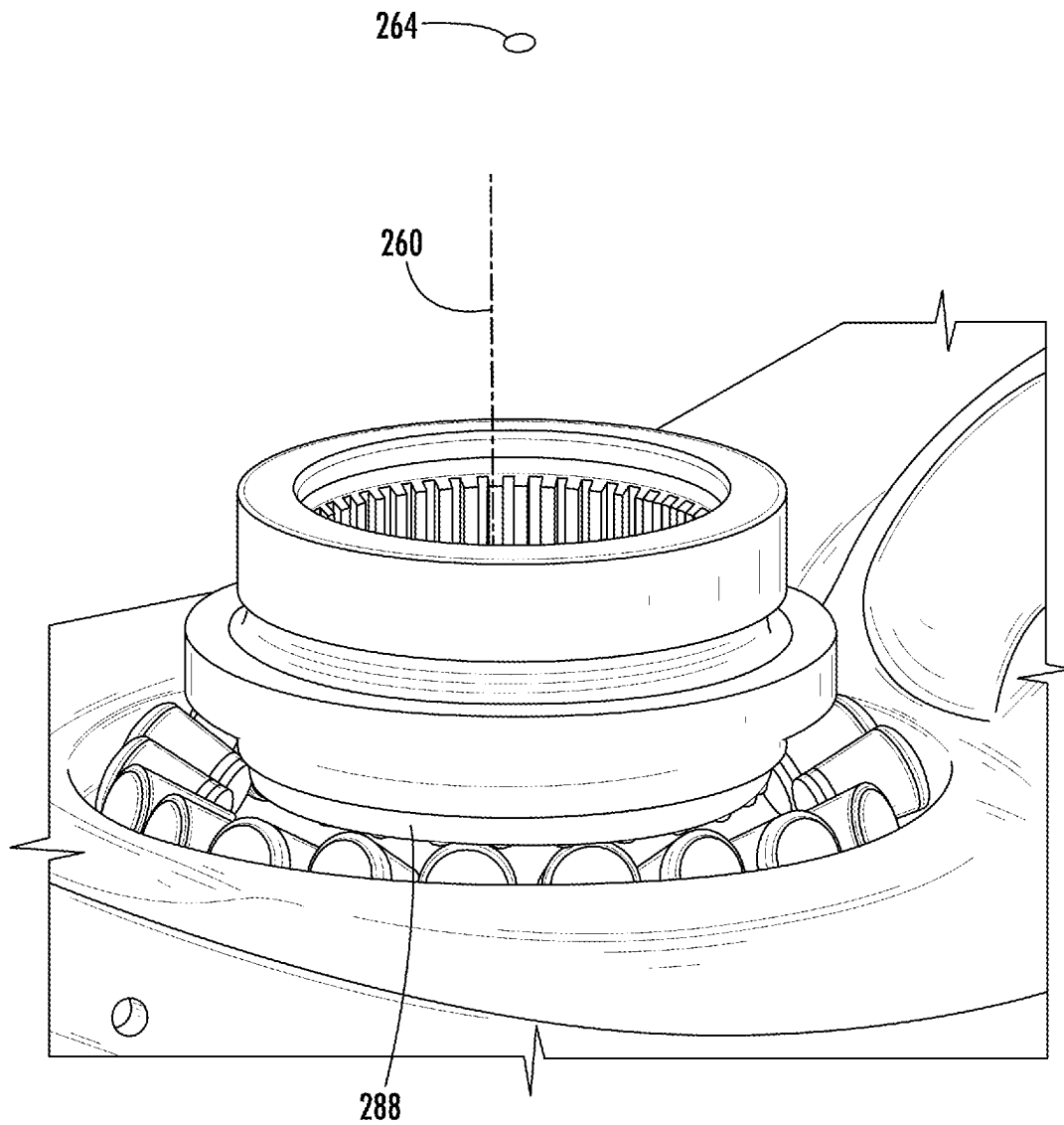


FIG. 20

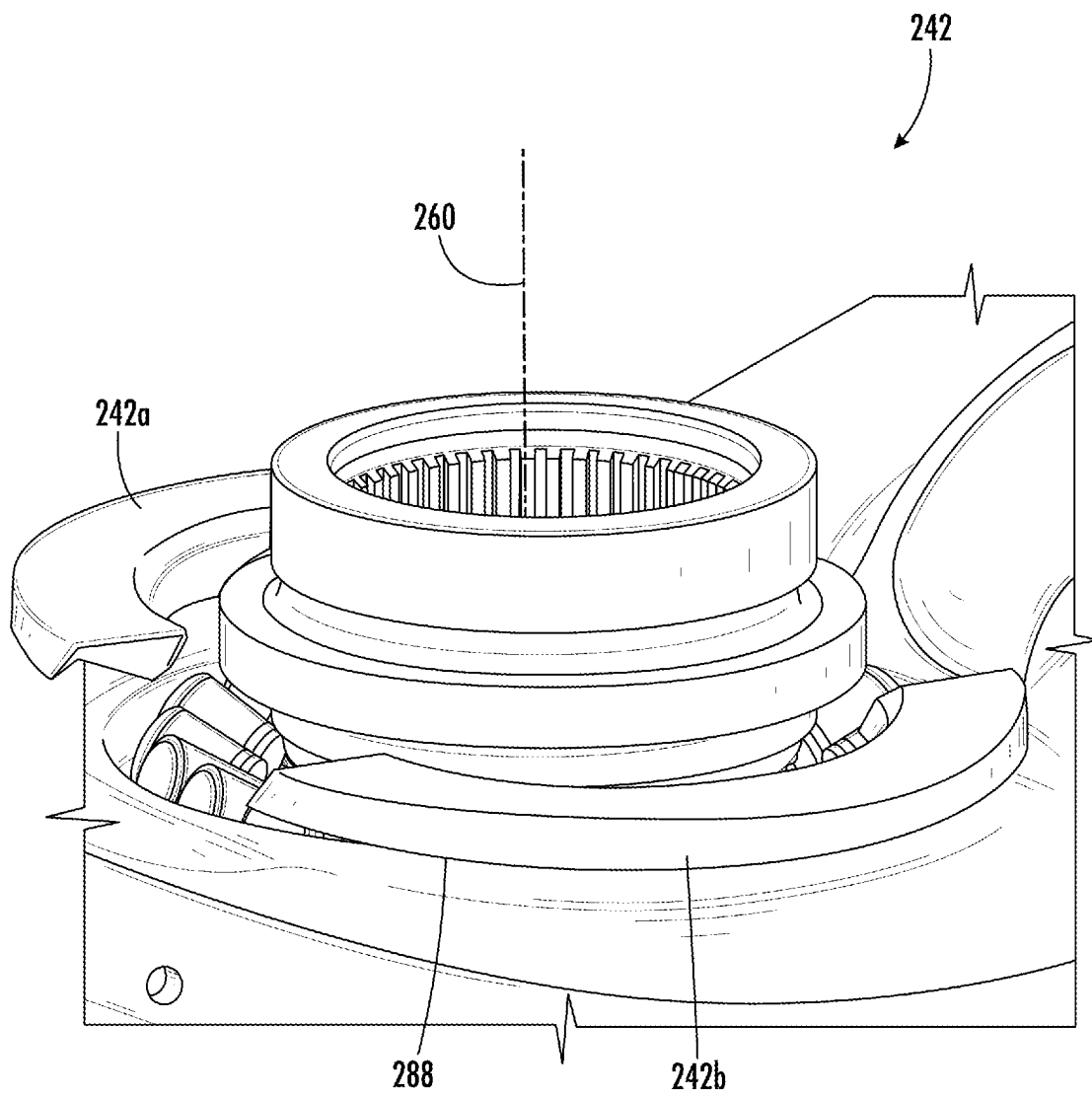


FIG. 21

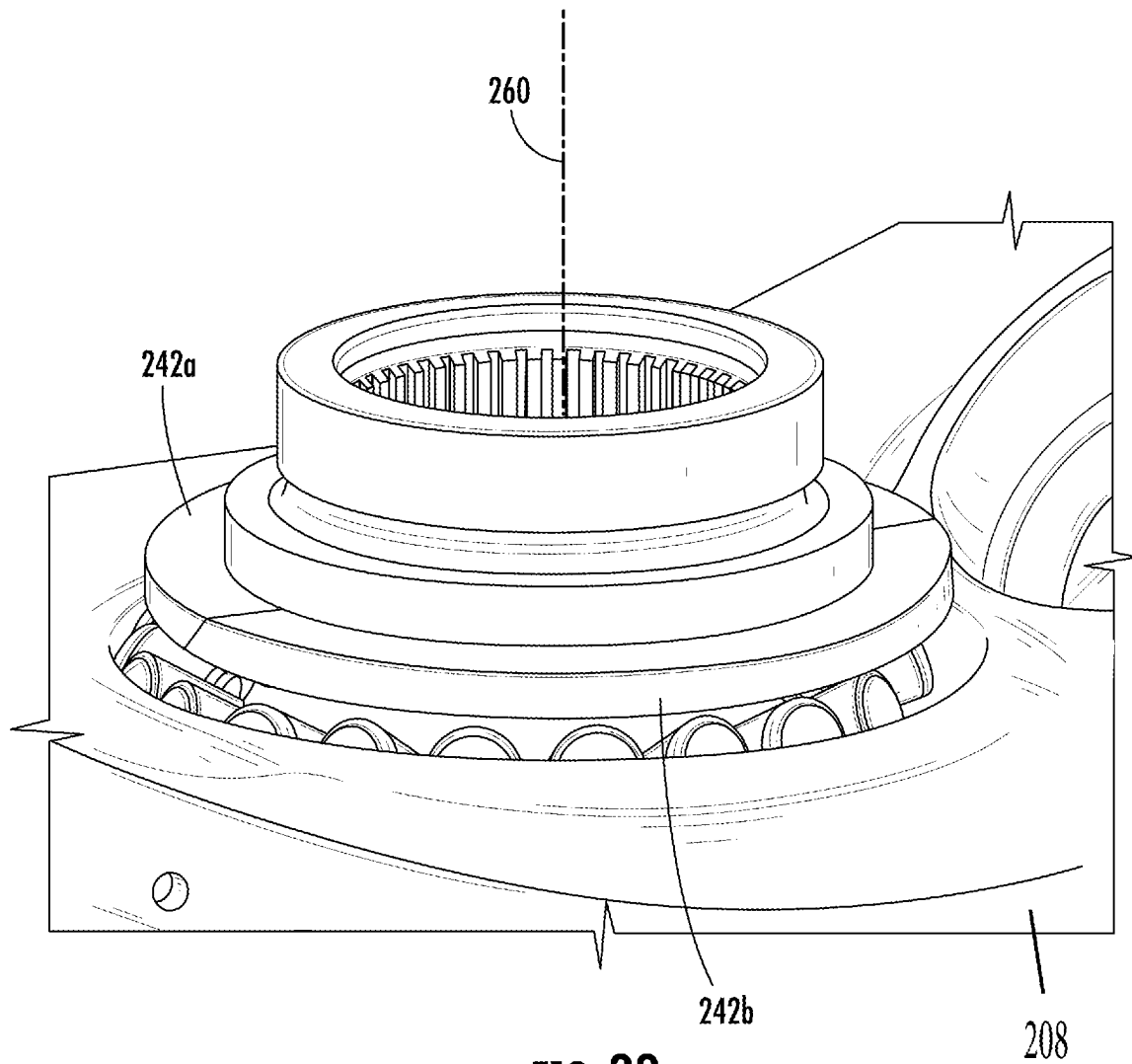


FIG. 22

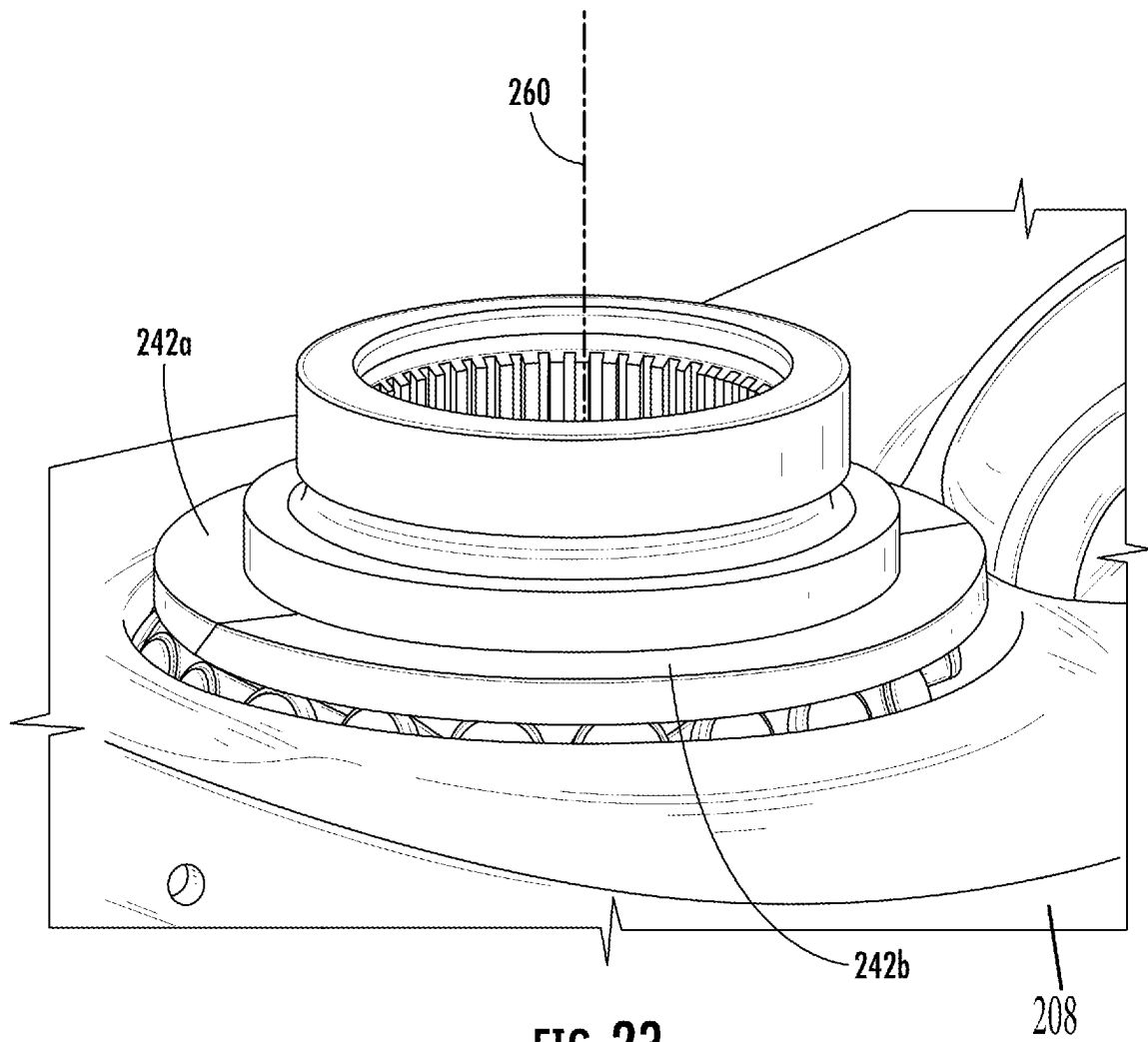


FIG. 23

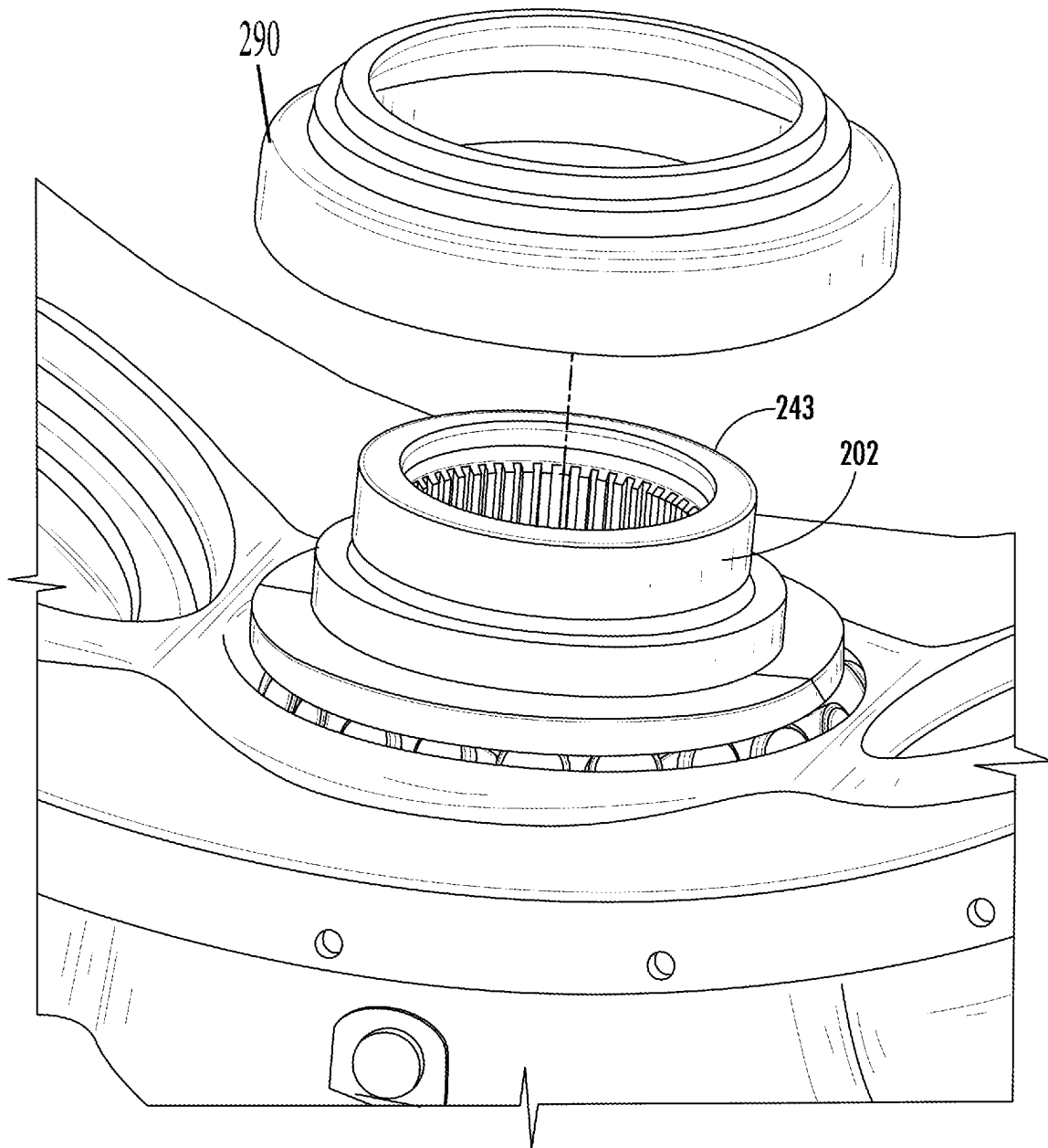


FIG. 24

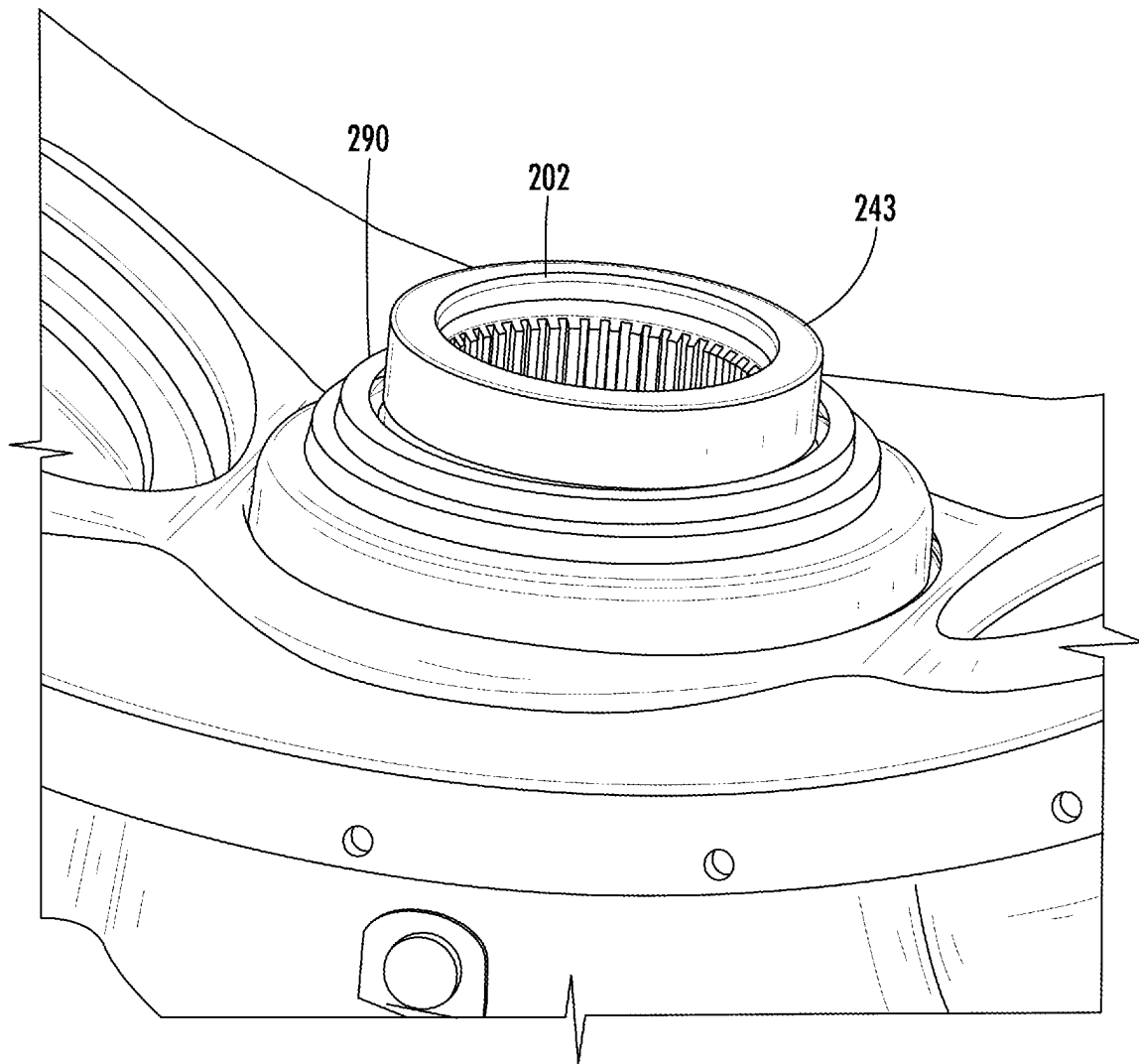


FIG. 25

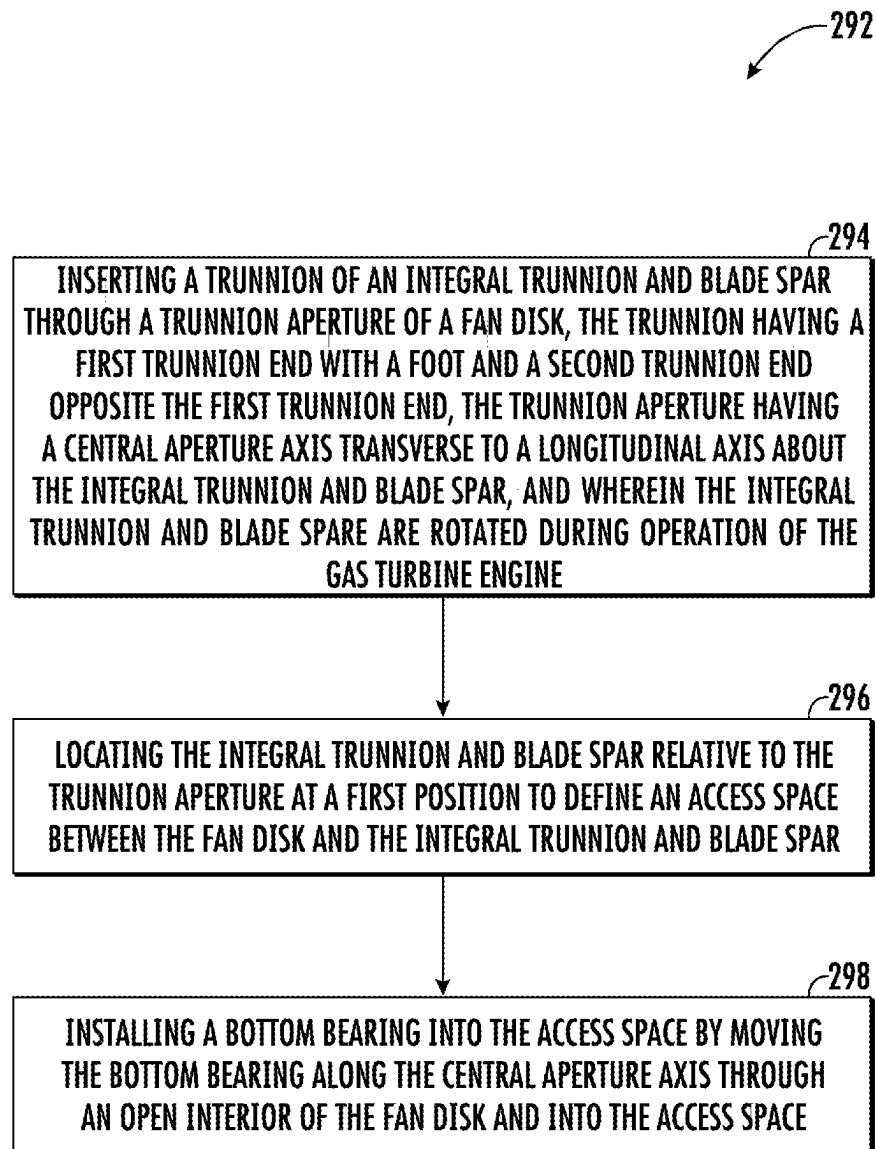


FIG. 26

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TRUNNION-TO-DISK CONNECTION FOR AN OPEN FAN CONFIGURATION AIRCRAFT POWERPLANT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Divisional Application of U.S. application Ser. No. 18/301,405 filed Apr. 17, 2023, which is hereby incorporated by reference in its entirety.

FIELD

The present disclosure relates to trunnion-to-disk connection for a turbofan engine and a turbofan engine including the same.

BACKGROUND

A gas turbine engine generally includes a turbomachine and a rotor assembly. Gas turbine engines, such as turbofan engines, may be used for aircraft propulsion. In the case of a turbofan engine, the rotor assembly may include a plurality of unducted rotor blades. It is desirable to provide a robust connection between each of the plurality of unducted rotor blades and a trunnion while also providing for ease of manufacturing. Improvements to trunnion-to-disk connections would be useful in the art.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the presently described technology, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

FIG. 1 is a cross-sectional view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.

FIG. 2 is a schematic of a prior art propeller hub for a propeller driven aircraft.

FIG. 3 is a schematic of a prior art support for a propeller connected to a propeller hub.

FIG. 4 is a schematic view of a trunnion-to-disk connection in accordance with an exemplary aspect of the present disclosure.

FIG. 5 is a schematic view of a fan disk in accordance with an exemplary aspect of the present disclosure.

FIG. 6 is a schematic view of a trunnion-to-disk connection in accordance with another exemplary aspect of the present disclosure.

FIG. 7 is a schematic view of a trunnion-to-disk connection in accordance with yet another exemplary aspect of the present disclosure.

FIG. 8 is a schematic view of a portion of the trunnion-to-disk connection of FIG. 7 illustrating a C-shaped seal in accordance with an exemplary aspect of the present disclosure.

FIG. 9 is a schematic view of a split insert prior to installation on a trunnion in accordance with an exemplary aspect of the present disclosure.

FIG. 10 is a schematic view of the split insert of FIG. 9 installed on the trunnion in accordance with an exemplary aspect of the present disclosure.

FIG. 11 is a schematic view of an upper bearing prior to installation on a trunnion in accordance with an exemplary aspect of the present disclosure.

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FIG. 12 is a schematic view of the upper bearing of FIG. 11 installed on a trunnion in accordance with an exemplary aspect of the present disclosure.

FIG. 13 is a schematic view of an insert prior to installation into a trunnion aperture of a fan disk in accordance with still another exemplary aspect of the present disclosure.

FIG. 14 is a schematic view of the insert of FIG. 13 upon installation into the trunnion aperture of the fan disk in accordance with still another exemplary aspect of the present disclosure.

FIG. 15 is a schematic view of a lower race of a bottom bearing prior to installation into a fan disk in accordance with still another exemplary aspect of the present disclosure.

FIG. 16 is a schematic view of a lower race of a bottom bearing depicted in FIG. 15 after installation into the fan disk in accordance with still another exemplary aspect of the present disclosure.

FIG. 17 is a schematic view of an integral trunnion and blade spar prior to installation into a trunnion aperture of a fan disk in accordance with still another exemplary aspect of the present disclosure.

FIG. 18 is a schematic view of an integral trunnion and blade spar of FIG. 17 inserted in the trunnion aperture of the fan disk in accordance with still another exemplary aspect of the present disclosure.

FIG. 19 is a schematic view of rolling elements of a bottom bearing prior to integration into in a bottom bearing of a trunnion-to-disk connection in accordance with still another exemplary aspect of the present disclosure.

FIG. 20 is a schematic view of rolling elements of the bottom bearing of FIG. 19 prior to installation of a lower bearing race in accordance with still another exemplary aspect of the present disclosure.

FIG. 21 is a schematic view of a split lower bearing race prior to installation in a bottom bearing in accordance with still another exemplary aspect of the present disclosure.

FIG. 22 is a schematic view of a split lower bearing race of FIG. 21 after installation in the bottom bearing in accordance with still another exemplary aspect of the present disclosure.

FIG. 23 is a schematic view of the assembly in FIG. 21 after pressurization of a hydraulic lift to engage a foot of a trunnion with a lower bearing race of a bottom bearing in accordance with still another exemplary aspect of the present disclosure.

FIG. 24 is a schematic view of a grease cap prior to installation to cover a lower bearing on a trunnion-to-disk connection in accordance with still another exemplary aspect of the present disclosure.

FIG. 25 is a schematic view of the grease cap of FIG. 24 after installation to cover the lower bearing on the trunnion-to-disk connection in accordance with still another exemplary aspect of the present disclosure.

FIG. 26 is a flow chart describing a method of attaching an integral trunnion and blade spar to a fan disk of a gas turbine engine in accordance with the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

The term “turbomachine” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.

The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

The terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a reference axis. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the reference axis. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the reference axis.

The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

A “third stream” as used herein means a non-primary air stream capable of increasing fluid energy to produce a minority of total propulsion system thrust. The third stream may generally receive inlet air (air from a ducted passage downstream of a primary fan) instead of freestream air (as the primary fan would). A pressure ratio of the third stream may be higher than that of the primary propulsion stream (e.g., a bypass or propeller driven propulsion stream). The thrust may be produced through a dedicated nozzle or through mixing of an airflow through the third stream with a primary propulsion stream or a core air stream, e.g., into a common nozzle.

As will be discussed in more detail below, the subject matter of the present disclosure is directed generally to connecting an integral trunnion and blade spar to a fan disk of an open fan configuration of a gas turbine engine. The integral trunnion and blade spar can be connected to the fan disk through a trunnion-to-disk connection which includes a

top bearing and a bottom bearing disposed between the fan disk and the trunnion of the integral trunnion and blade spar. A cavity can be provided between the trunnion and fan disk and between the top bearing and bottom bearing. A hydraulic lift can be provided to supply hydraulic fluid to the cavity and cause relative movement and/or stress to develop between the trunnion and the fan disk. A foot of the trunnion can engage a lower race of a bottom bearing upon pressurization of the cavity by the hydraulic lift. During an installation process the integral trunnion and blade spar can be inserted such that a second end of the trunnion extends past the open fan disk to provide sufficient space to install the bottom bearing. The open disk includes a large open interior sufficient to install rolling elements of the lower bearing in an axial direction with respect to a central aperture axis which, in some forms, is transverse to a longitudinal axis of the gas turbine engine. After installation of the bottom bearing the hydraulic lift can be activated to pressurize the cavity and preload the bearings, at which point a spanner nut can be tightened to lock the assembly in place. It is envisioned herein that the trunnion-to-disk connection can be applied to a high blade count, high radius ratio open fan (e.g., an unducted single fan) engine. Unducted single fan architectures require higher blade solidity relative to existing turboprop applications to enable high velocity cruise speeds. Higher blade counts require higher radius ratio fan disks. Unducted single fan designs, such as those in which integral trunnion and blade spars disclosed herein may be connected to a fan disk, may have blade counts greater than 10 and radius ratios greater than 0.200. Vibratory blade loading may be higher than existing turboprop applications due to higher flight velocities. As a result, larger bearings and bearing spacing are required to maintain contact stress requirements.

Referring now to FIG. 1, a schematic cross-sectional view of a gas turbine engine **100**, also referred to herein as “engine **100**”, is provided according to an example embodiment of the present disclosure. Particularly, FIG. 1 provides a turbofan engine having a rotor assembly with a single stage of unducted rotor blades. In such a manner, the rotor assembly may be referred to herein as an “unducted fan,” or the entire engine **100** may be referred to as a “turbofan engine” or an “unducted turbofan engine.” In addition, the engine **100** of FIG. 1 includes a third stream extending from the compressor section to a rotor assembly flowpath over the turbomachine, as will be explained in more detail below.

For reference, the engine **100** defines an axial direction A, a radial direction R, and a circumferential direction C. Moreover, the engine **100** defines an axial centerline or longitudinal axis **112** that extends along the axial direction A. In general, the axial direction A extends parallel to the longitudinal axis **112**, the radial direction R extends outward from and inward to the longitudinal axis **112** in a direction orthogonal to the axial direction A, and the circumferential direction extends three hundred sixty degrees (360°) around the longitudinal axis **112**. The engine **100** extends between a forward end **114** and an aft end **116**, e.g., along the axial direction A.

The engine **100** includes a turbomachine **120** and a rotor assembly, also referred to a fan section **150**, positioned upstream thereof. Generally, the turbomachine **120** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. Particularly, as shown in FIG. 1, the turbomachine **120** includes a core cowl **122** that defines an annular core inlet **124**. The core cowl **122** further encloses at least in part a low pressure system and a high pressure system. For example, the core cowl **122** depicted encloses and supports at least in part a

booster or low pressure (“LP”) compressor **126** for pressurizing the air that enters the turbomachine **120** through core inlet **124**. A high pressure (“HP”), multi-stage, axial-flow compressor **128** receives pressurized air from the LP compressor **126** and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor **130** of the combustion section where fuel is injected into the pressurized air stream and ignited to raise the temperature and energy level of the pressurized air.

It will be appreciated that as used herein, the terms “high/low speed” and “high/low pressure” are used with respect to the high pressure/high speed system and low pressure/low speed system interchangeably. Further, it will be appreciated that the terms “high” and “low” are used in this same context to distinguish the two systems, and are not meant to imply any absolute speed and/or pressure values.

The high energy combustion products flow from the combustor **130** downstream to a high pressure turbine **132**. The high pressure turbine **132** drives the high pressure compressor **128** through a high pressure shaft **136**. In this regard, the high pressure turbine **132** is drivingly coupled with the high pressure compressor **128**. The high energy combustion products then flow to a low pressure turbine **134**. The low pressure turbine **134** drives the low pressure compressor **126** and components of the fan section **150** through a low pressure shaft **138**. In this regard, the low pressure turbine **134** is drivingly coupled with the low pressure compressor **126** and components of the fan section **150**. The LP shaft **138** is coaxial with the HP shaft **136** in this example embodiment. After driving each of the turbines **132**, **134**, the combustion products exit the turbomachine **120** through a turbomachine exhaust nozzle **140**.

Accordingly, the turbomachine **120** defines a working gas flowpath or core duct **142** that extends between the core inlet **124** and the turbomachine exhaust nozzle **140**. The core duct **142** is an annular duct positioned generally inward of the core cowl **122** along the radial direction R. The core duct **142** (e.g., the working gas flowpath through the turbomachine **120**) may be referred to as a second stream.

The fan section **150** includes a fan **152**, which is the primary fan in this example embodiment. For the depicted embodiment of FIG. 1, the fan **152** is an open rotor or unducted fan **152**. In such a manner, the engine **100** may be referred to as an open rotor or open fan engine.

As depicted, the fan **152** includes an array of fan blades **154** (only one shown in FIG. 1). The fan blades **154** are rotatable, e.g., about the longitudinal axis **112**. As noted above, the fan **152** is drivingly coupled with the low pressure turbine **134** via the LP shaft **138**. For the embodiments shown in FIG. 1, the fan **152** is coupled with the LP shaft **138** via a speed reduction gearbox **155**, e.g., in an indirect-drive or geared-drive configuration.

Moreover, the array of fan blades **154** can be arranged in equal spacing around the longitudinal axis **112**. Each fan blade **154** has a root and a tip and a span defined therebetween, and more specifically defines a tip radius R_{TIP} from the longitudinal axis **112** to the tips of the fan blades **154** along the radial direction R. Each fan blade **154** defines a central blade axis **156**. For this embodiment, each fan blade **154** of the fan **152** is rotatable about its central blade axis **156**, e.g., in unison with one another. One or more actuators **158** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan blades **154** about their respective central blades’ axes **156**.

The fan section **150** further includes an outlet guide vane array **160** that includes outlet guide vanes **162** (only one shown in FIG. 1; sometimes also referred to as fan guide

vanes) disposed around the longitudinal axis **112**. For this embodiment, the outlet guide vanes **162** are not rotatable about the longitudinal axis **112**. Each outlet guide vane **162** has a root and a tip and a span defined therebetween. The outlet guide vanes **162** may be unshrouded as shown in FIG. 1 or, alternatively, may be shrouded, e.g., by an annular shroud spaced outward from the tips of the outlet guide vanes **162** along the radial direction R or attached to the outlet guide vanes **162**.

As will be appreciated, the outlet guide vanes **162** each define an outlet guide vane (OGV) span **164** along the radial direction R from a root to a tip. Additionally, the outlet guide vanes **162** are spaced from the fan blade **154** along the axial direction A by a distance or spacing **166**. The spacing **166** is measured from an aft-most edge of the fan blade **154** to a forward-most edge of the outlet guide vanes **162** along the axial direction A.

In the embodiment depicted, as noted above, each outlet guide vane **162** is configured as a fixed guide vane, unable to be pitched about a central blade axis. The outlet guide vanes **162** are thus mounted to a fan cowl **170** in a fixed manner.

It will be appreciated, however, that in other embodiments, the outlet guide vanes **162** may alternatively be variable pitch outlet guide vanes **162**.

As shown in FIG. 1, in addition to the fan **152**, which is unducted, a ducted fan **184** is included aft of the fan **152**, such that the engine **100** includes both a ducted and an unducted fan which both serve to generate thrust through the movement of air without passage through at least a portion of the turbomachine **120** (e.g., without passage through the HP compressor **128** and combustion section for the embodiment depicted). The ducted fan **184** is rotatable about the same axis (e.g., the longitudinal axis **112**) as the fan blade **154**. The ducted fan **184** is, for the embodiment depicted, driven by the low pressure turbine **134** (e.g. coupled to the LP shaft **138**). In the embodiment depicted, as noted above, the fan **152** may be referred to as the primary fan, and the ducted fan **184** may be referred to as a secondary fan. It will be appreciated that these terms “primary” and “secondary” are terms of convenience, and do not imply any particular importance, power, or the like.

The ducted fan **184** includes a plurality of fan blades (not separately labeled in FIG. 1) arranged in a single stage, such that the ducted fan **184** may be referred to as a single stage fan. The fan blades of the ducted fan **184** can be arranged in equal spacing around the longitudinal axis **112**. Each blade of the ducted fan **184** has a root and a tip and a span defined therebetween.

The fan cowl **170** annularly encases at least a portion of the core cowl **122** and is generally positioned outward of at least a portion of the core cowl **122** along the radial direction R. Particularly, a downstream section of the fan cowl **170** extends over a forward portion of the core cowl **122** to define a fan duct flowpath, or simply a fan duct **172**. According to this embodiment, the fan flowpath or fan duct **172** may be understood as forming at least a portion of the third stream of the engine **100**.

Incoming air may enter through the fan duct **172** through a fan duct inlet **176** and may exit through a fan exhaust nozzle **178** to produce propulsive thrust. The fan duct **172** is an annular duct positioned generally outward of the core duct **142** along the radial direction R. The fan cowl **170** and the core cowl **122** are connected together and supported by a plurality of substantially radially-extending, circumferentially-spaced stationary struts **174** (only one shown in FIG. 1). The stationary struts **174** may each be aerodynamically

contoured to direct air flowing thereby. Other struts in addition to the stationary struts **174** may be used to connect and support the fan cowl **170** and/or core cowl **122**. In many embodiments, the fan duct **172** and the core duct **142** may at least partially co-extend (generally axially) on opposite sides (e.g., opposite radial sides) of the core cowl **122**. For example, the fan duct **172** and the core duct **142** may each extend directly from a leading edge **144** of the core cowl **122** and may partially co-extend generally axially on opposite radial sides of the core cowl **122**.

The engine **100** also defines or includes an inlet duct **180**. The inlet duct **180** extends between an engine inlet **182** and the core inlet **124**/fan duct inlet **176**. The engine inlet **182** is defined generally at the forward end of the fan cowl **170** and is positioned between the fan **152** and the outlet guide vane array **160** along the axial direction A. The inlet duct **180** is an annular duct that is positioned inward of the fan cowl **170** along the radial direction R. Air flowing downstream along the inlet duct **180** is split, not necessarily evenly, into the core duct **142** and the fan duct **172** by a fan duct splitter or leading edge **144** of the core cowl **122**. In the embodiment depicted, the inlet duct **180** is wider than the core duct **142** along the radial direction R. The inlet duct **180** is also wider than the fan duct **172** along the radial direction R.

Notably, for the embodiment depicted, the engine **100** includes one or more features to increase an efficiency of a third stream thrust, $F_{n_{3S}}$ (e.g., a thrust generated by an airflow through the fan duct **172** exiting through the fan exhaust nozzle **178**, generated at least in part by the ducted fan **184**). In particular, the engine **100** further includes an array of inlet guide vanes **186** positioned in the inlet duct **180** upstream of the ducted fan **184** and downstream of the engine inlet **182**. The array of inlet guide vanes **186** are arranged around the longitudinal axis **112**. For this embodiment, the inlet guide vanes **186** are not rotatable about the longitudinal axis **112**. Each inlet guide vane **186** defines a central blade axis (not labeled for clarity), and is rotatable about its respective central blade axis, e.g., in unison with one another. In such a manner, the inlet guide vanes **186** may be considered a variable geometry component. One or more actuators **188** are provided to facilitate such rotation and therefore may be used to change a pitch of the inlet guide vanes **186** about their respective central blade axes. However, in other embodiments, each inlet guide vane **186** may be fixed or unable to be pitched about its central blade axis.

Further, located downstream of the ducted fan **184** and upstream of the fan duct inlet **176**, the engine **100** includes an array of outlet guide vanes **190**. As with the array of inlet guide vanes **186**, the array of outlet guide vanes **190** are not rotatable about the longitudinal axis **112**. However, for the embodiment depicted, unlike the array of inlet guide vanes **186**, the array of outlet guide vanes **190** are configured as fixed-pitch outlet guide vanes.

Further, it will be appreciated that for the embodiment depicted, the fan exhaust nozzle **178** of the fan duct **172** is further configured as a variable geometry exhaust nozzle. In such a manner, the engine **100** includes one or more actuators **192** for modulating the variable geometry exhaust nozzle. For example, the variable geometry exhaust nozzle may be configured to vary a total cross-sectional area (e.g., an area of the nozzle in a plane perpendicular to the longitudinal axis **112**) to modulate an amount of thrust generated based on one or more engine operating conditions (e.g., temperature, pressure, mass flowrate, etc. of an airflow through the fan duct **172**). A fixed geometry exhaust nozzle may also be adopted.

The combination of the array of inlet guide vanes **186** located upstream of the ducted fan **184**, the array of outlet guide vanes **190** located downstream of the ducted fan **184**, and the fan exhaust nozzle **178** may result in a more efficient generation of third stream thrust, $F_{n_{3S}}$, during one or more engine operating conditions. Further, by introducing a variability in the geometry of the inlet guide vanes **186** and the fan exhaust nozzle **178**, the engine **100** may be capable of generating more efficient third stream thrust, $F_{n_{3S}}$, across a relatively wide array of engine operating conditions, including takeoff and climb (where a maximum total engine thrust $F_{n_{Total}}$ is generally needed) as well as cruise (where a lesser amount of total engine thrust, $F_{n_{Total}}$, is generally needed).

Moreover, referring still to FIG. 1, in exemplary embodiments, air passing through the fan duct **172** may be relatively cooler (e.g., lower temperature) than one or more fluids utilized in the turbomachine **120**. In this way, one or more heat exchangers **198** may be positioned in thermal communication with the fan duct **172**. For example, one or more heat exchangers **198** may be disposed within the fan duct **172** and utilized to cool one or more fluids from the core engine with the air passing through the fan duct **172**, as a resource for removing heat from a fluid, e.g., compressor bleed air, oil or fuel.

Although not depicted, the heat exchanger **198** may be an annular heat exchanger extending substantially 360 degrees in the fan duct **172** (e.g., at least 300 degrees, such as at least 330 degrees). In such a manner, the heat exchanger **198** may effectively utilize the air passing through the fan duct **172** to cool one or more systems of the engine **100** (e.g., lubrication oil systems, compressor bleed air, electrical components, etc.). The heat exchanger **198** uses the air passing through duct **172** as a heat sink and correspondingly increases the temperature of the air downstream of the heat exchanger **198** and exiting the fan exhaust nozzle **178**.

It will be appreciated, that for the purposes of discussion in the present disclosure, the ducted fan **184**, the fan cowl **170**, the inlet duct **180**, and the fan duct **172** may all be considered part of the turbomachine **120**.

It will be appreciated that the exemplary turbofan engine **100** depicted in FIG. 1 is provided by way of example only, and that in other embodiments, the turbofan engine **100** may have any other suitable configuration. For example, in other embodiments, the turbofan engine **100** may not include the fan duct **172**/third stream, and as such may be configured as a "two stream" engine. Additionally, or alternatively, in other embodiments, the turbofan engine **100** may be configured as a direct drive engine (i.e., without the gearbox **155**), as a fixed-pitch engine (i.e., without the pitch change mechanism/actuator(s) **158**), etc.

The rotor assembly can also include a spinner cone **210** fastened on a forward end of the rotor assembly to provide for an aerodynamic flow path required to reach the radial location of the engine inlet **182**.

With particular respect to the fan blades **154** of the rotor assembly discussed above, in one exemplary form the fan blades **154** include an integral trunnion and blade spar **200** having a trunnion **202** and blade spar **204** upon which is coupled a fan airfoil **206** through a physical attachment or bonding process. The integral trunnion and blade spar **200** can be made as an integral component through any suitable manufacturing process, including but not limited to any suitable bonding process, such as through metallurgical bonding, or casting process, or physical attachment process, to set forth just a few non-limiting examples. In one form the trunnion **202** is integral with the blade spar **200** where both

are of metallic material in one form, or metallic/polymatrix composite (PMC) hybrid in another form. Other material types are also contemplated.

The trunnion **202** of the integral trunnion and blade spar **200** is coupled to a fan disk **208** that is driven by the LP shaft **138**. The connection point of the LP shaft **138** to the fan disk **208** is indicated in FIG. 1 as being located on an axial forward end of the fan disk **208**, but in other embodiments, the LP shaft **138** can be connected to the fan disk **208** in other locations, including on an axial aft end of the fan disk **208**. The coupled connection between the trunnion **202** of the integral trunnion and blade spar **200** will be described further below.

FIGS. 2 and 3 depict a prior art example of a propeller driven engine which includes an assembly used to couple propeller blades **212** to a propeller hub **214**. Each of the propellers extend along a propeller axis **218**, and the entire propeller assembly rotates about an engine axis **220**. The propeller blades **212** can be integral with a trunnion **216** and are coupled via the trunnion **216** to the propeller hub **214** as depicted in FIG. 3. A similar coupling is also depicted in U.S. Pat. No. 4,921,403. Of note in FIG. 2, the hub radially extends substantially to the engine axis **220**. Also, of note in FIG. 2, a rolling element required for the lower bearing must be installed via a bearing port **222** in an axial direction along the engine axis **220** through a wall of the propeller hub **214**. After final installation the propeller hub **214** is flooded with a lubricant, such as grease, which is shared among the lower bearings for each of the separate trunnions associated with the propeller blades. During operation, the lubricant is flung radially outward and falls back after engine operation. Existing turboprop applications in which the construction of FIGS. 2 and 3 may be used do not require high blade solidity due to lower cruise velocities. Lower blade counts enable lower radius ratio fan disks. Turboprop designs, such as those that may use the constructions of FIGS. 2 and 3, may have blade counts less than 10 and radius ratios less than 0.200.

Turning now to FIG. 4, the trunnion **202** is coupled to the fan disk **208** through a trunnion-to-disk connection **224** which includes a top bearing **226**, a bottom bearing **228**, and a hydraulic lift **230**. As will be appreciated, the top bearing **226** and bottom bearing **228** aid in positioning and supporting the trunnion **202** within the fan disk **208**, but also crucially permit relative motion to occur between the trunnion **202** and the fan disk **208**. The top bearing **226** can be used in some forms to provide a wheelbase to react a moment during operation of the fan blades **154**. The bottom bearing **228** can be used in some forms to provide a primary radial retention of the fan blades **154**. During an assembly process, the hydraulic lift **230** is used to provide translation of the trunnion **202** relative to the fan disk **208** as will be described further below.

The top bearing **226** includes an upper race **232**, a rolling element **234** (shown without cross hatching in FIG. 4 to indicate an out-of-plane view), and a lower race **236**. The upper race **232** is press fit or otherwise connected with the trunnion **202** and can rotate and/or translate with the trunnion **202** during operation of the fan blades **154** (FIG. 1) and/or installation of the trunnion **202** with the fan disk **208**. In the illustrated embodiment, the rolling elements **234** are depicted as roller bearings, but other embodiments can incorporate different types of rolling elements, such as, but not limited to, ball or spherical bearings. Other embodiments disclosed herein illustrate other bearing types. The lower race **236** is either captured by or serves as a bearing carrier capable of translating with the trunnion **202** during an

installation procedure of the integral trunnion and blade spar **204** to the fan disk **208**. In some forms, the lower race **236** can be connected with the fan disk **208** through a pin, pilot feature, etc. which can serve to rotatably lock the lower race **236** to the fan disk **208**. A lubricant can be provided within the top bearing **226**. The top bearing **226** is located near a first end **237** of the trunnion **202**, where such first end **237** will be distinguished from another end of the trunnion **202** in which the bottom bearing **228** is located.

The bottom bearing **228** is separable (e.g., separate components can be manipulated and installed separately) and includes an upper race **238**, rolling element **240**, and a lower race **242**. The upper race **238** is press fit or otherwise connected with the fan disk **208**, such as via an overhang **239** formed in the fan disk **208**. In some forms, the upper race **238** can be connected with the fan disk **208** through a pin, pilot feature, etc. which can serve to rotatably lock the upper race **238** to the fan disk **208**. In the illustrated embodiment the rolling elements **240** are depicted as tapered roller bearings, but other embodiments can incorporate different types of rolling elements, such as, but not limited to, ball or spherical bearings. Other embodiments disclosed herein illustrate other bearing types. The lower race **242** is press fit or otherwise connected with the trunnion **202** and can rotate and/or translate with the trunnion during operation of the fan blades **154** (FIG. 1) and/or installation of the trunnion **202** with the fan disk **208**. The lower race **242** of the bottom bearing **228** abuts a bearing contact surface **235** of a foot **241** of the trunnion **202** which can provide support to the lower race **242** over all or a portion of the width of the lower race **242**. A lubricant can be provided within the top bearing **226**. The bottom bearing **228** is located near a second end **243** of the trunnion **202**, where the second end **243** is opposite the first end **237**. In one form the top bearing **226** and the bottom bearing **228** are located between the first end **237** and the second end **243**.

The hydraulic lift **230** includes a hydraulic port **244** to which a suitable hydraulic fitting can be attached to deliver a hydraulic material, such as, but not limited to, grease, through the hydraulic port **244** and into a cavity **246**. The hydraulic port **244** and the cavity **246** are structured to receive a high pressure hydraulic fluid to cause movement and/or impart stress between the trunnion **202** and the fan disk **208**. The cavity **246** is defined between the lower race **236** of the top bearing **226** and the upper race **238** of the bottom bearing **228**, and is also defined between an inner wall of the fan disk **208** and the outer surface of the trunnion **202**. Prior to pressurization, the trunnion **202** can be translated to change the height of the cavity **246** since the top bearing **226** and trunnion **202** are permitted to slide in place. For example, when the cavity **246** is under no pressure from a hydraulic fluid, the trunnion **202** can be translated to collapse the cavity **246** to a minimum height. When the cavity **246** is under pressure, however, the trunnion **202** can be translated to expand the cavity **246** to a maximum height as will be discussed further below. When pressurized hydraulic fluid is received within cavity **246** such pressure compresses the top bearing **226** but also urges the lower race **236** of the top bearing **226** to translate in a direction from the second end **243** to the first end **237** and thereby compresses the bottom bearing **228** between the foot **241** of the trunnion **202** and the overhang **239** of the fan disk **208**. Pressurizing the cavity **246** permits the top bearing **226** and the bottom bearing **228** to be preloaded.

A spanner nut **248** can be affixed to lock the trunnion **202** in place. In one form the spanner nut **248** is a threaded nut affixed to a bearing carrier that forms the upper race **232** of

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the top bearing 226 as shown in the embodiment depicted in FIG. 4. Other intermediate structure can also be used. When the cavity 246 of the hydraulic lift 230 is fully pressurized and the top bearing 226 and bottom bearing 228 are in a preloaded state, the spanner nut 248 can be threaded into place. In some embodiments, the spanner nut 248 can be hand tightened, but in other embodiments, the spanner nut 248 can be torqued into place through appropriate tooling. Once the spanner nut 248 is in place the hydraulic fitting connected to the hydraulic port 244 can be disconnected and a port cover inserted to close off the hydraulic port 244.

A number of inserts can be used to occupy space defined between the lower race 236 of the top bearing 226 and the upper race 238 of the bottom bearing 228, and also defined between an inner wall of the fan disk 208 and the outer surface of the trunnion 202. Such inserts can be connected to either of the trunnion 202 or the fan disk 208. Depicted in FIG. 4 are inserts 250 and 252. The inserts 250 and 252 can take a variety of forms including, but not limited to, foam inserts. In one form, the foam inserts 250 and 252 can be a closed foam construction formed from any suitable material, including but not limited to foamed metal. The foam inserts 250 and 252 can be used to decrease the volume needed when supplying hydraulic fluid to urge the trunnion 202 to move and therefore create loading upon the top bearing 226 and the bottom bearing 228.

The construction depicted in FIG. 4 includes the use of several sealing elements to prevent intrusion of foreign materials and/or leakage of lubricant. O-rings 254 can be used to seal between various stationary and moving parts. For example, an O-ring 254 is located between the relatively stationary foam insert 250 and an inner surface of the fan disk 208. An O-ring 254 is also located between the foam insert 252 and the bearing carrier of the top bearing 226, where such O-ring permits sliding movement between the bearing carrier of the top bearing 226 and the foam insert 250. A C-shaped seal 256 can be used in locations to aid in sealing between parts that have relative movement. An example of a C-shaped seal is illustrated further below in FIG. 8. It will be appreciated that other types of seals can be used besides O-rings 254 and C-shaped seals 256. In fact, in some forms any of the O-rings 254 can be used in place of any C-shaped seals 256, and vice versa, depending on the application.

The lower race 236 of the top bearing 226 includes not only the race for the rolling elements 234, but also includes an upturned leg 257, a base 259, and a downturned leg 261. The upturned leg 257 extending in the direction of travel of the trunnion 202 when the cavity is pressurized abuts an inner surface of the fan disk 208, which interface also includes a seal in the form of an O-ring 254. A bottom facing surface of the base 259 defines a top portion of the cavity 246. The downturned leg 261 is oriented opposite the upturned leg 257 relative to the base 259 and is located between the insert 250 and insert 252. The downturned leg 261 also defines a wall of the cavity 246. Although only a single seal is depicted between the downturned leg 261 and the insert 250 owing to the relative movement between those parts when the trunnion 202 is moved as a result of pressurizing the cavity 246, another seal could also be used between the downturned leg 261 and the insert 252.

Referencing now both FIGS. 4 and 5, the fan disk 208 includes several features including a plurality of trunnion apertures 258 each of which is constructed to receive an associated integral trunnion and blade spar 200 such as that illustrated in FIG. 4. Each of the trunnion apertures 258 is arranged along a central aperture axis 260 along which the

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trunnion 202 is inserted. When inserted, the trunnion 202 can be rotated about the central aperture axis 260 to rotate the fan blades 154, where such rotation can be accomplished by a bell crank 262 attached to the trunnion 202. An actuator (e.g., 158 from FIG. 1) can be used to manipulate the orientation of the bell crank 262, and therefore the orientation of the fan blades 154. It will be appreciated that the central blade axis 156 (FIG. 1) can be coincident with the central aperture axis 260 in some embodiments, but in other embodiments need not be coincident. It will also be appreciated that the central blade axis 156 and central aperture axis 260 are substantially transverse to the longitudinal axis 112 (FIG. 1). In some forms either or both of the central blade axis 156 and central aperture axis 260 are perpendicular to the longitudinal axis 112.

The fan disk 208 is annular in shape and includes several provisions to incorporate it into the gas turbine engine 100 (FIG. 1). The annular shape includes a large open interior 264 within which an installation procedure of the bottom bearing 228 can be performed as discussed further below. The fan disk 208 is constructed to rotate about the longitudinal axis 112 when installed on the gas turbine engine 100. A forward end 266 can be coupled with the spinner cone 210, and a rear end 268 to the LP shaft 138 via a plurality of fasteners inserted through respective ones of a plurality of fastener apertures 270. Various fastening techniques are contemplated for both of the forward end 266 to the spinner cone 210 and the rear end 268 to the LP shaft 138. Furthermore, in some embodiments, additional components, structural or otherwise, may be coupled between the forward end 266 and spinner cone 210 as well as between the rear end 268 and the LP shaft 138.

The fan disk 208 also includes other several provisions to aspects of the trunnion-to-disk connection 224. For example, the fan disk 208 includes a lubricant passage 272 which is sized to receive an insert for the hydraulic port 244. The fan disk 208 also includes a spanner nut bearing surface 274 sized to support and bear against the spanner nut 248 to provide preloading on the top bearing 226 and bottom bearing 228. The fan disk 208 also includes a disk bearing surface in the form of overhang 239 sized to support and bear against the upper race 238 of the bottom bearing 228. The fan disk 208 also includes a portion of the inner surface designated at 275 in FIG. 4 which indicates the minimum radius of the fan disk 208 at the location aligned with the trunnion aperture 258. The bearing contact surface 235 is at a radius from the longitudinal axis 112 that is less than the minimum radius of the fan disk 208 from the longitudinal axis 112 at that portion of the inner surface designated at 275 which represents the location aligned with the trunnion aperture 258. The fan disk 208 can include an axial forward end having a minimum radius 277 from the longitudinal axis 112 and an axial aft end with a minimum radius 279 from the longitudinal axis 112. The radius of the bearing contact surface 235 from the longitudinal axis 112 can, but need not, be less than either or both of the minimum radii 277 and 279.

Turning now to FIG. 6, and with reference to FIG. 4, another embodiment of the trunnion-to-disk connection 224 is illustrated. Like reference numerals refer to like elements. Differences in the embodiment illustrated in FIG. 6 include a shortened downturned leg 261 of the lower race 236 of the top bearing 226. Instead of extending the downturned leg 261 to a location between the insert 250 and 252, an additional component 278 is inserted having a first portion located between the inserts 250 and 252, and a second portion extending axially away from the first portion that abuts the downturned leg 261 on a radially outer side of the

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downturned leg **261** (where ‘radially’ in this sense is measured from the central aperture axis **260**). Seals are provided on either side of the first portion which abuts the insert **250** and insert **252**. The bottom bearing **228** also includes one portion of the bearing race formed in the insert **252**. The rolling elements **234** and **240** are ball and/or spherical bearings. The cavity **246** in FIG. 6 is therefore defined between the base **259** (see FIG. 4) of the lower race **236**, a surface of the component **278**, and a top surface of the insert **250**.

Turning now to FIG. 7, and with reference to FIG. 4, another embodiment of the trunnion-to-disk connection **224** is illustrated. Like reference numerals refer to like elements. Differences in the embodiment depicted in FIG. 7 include the absence of a base **259** and a downturned leg **261** in the lower race **236** of the top bearing **226**. The absence of the base **259** and the downturned leg **261** of the lower race **236** provides a greater flexibility in the lower race **236** as compared to a lower race **236** having the base **259** and downturned leg **261**. Instead of the base **259** of the lower race **236** defining a top part of the cavity **246**, a sleeve **280** is positioned such that a lateral extension of the sleeve defines a top part of the cavity **246**, and a lower extending arm defines a lateral wall of the cavity **246** opposite the hydraulic port **244**. A shelf of the fan disk **208** forms a bottom part of the cavity **246**. The lower extending arm of the sleeve **280** is positioned between the insert **252** and an inner surface of the fan disk **208**.

FIG. 8 depicts a portion of the structure shown in FIG. 7, and specifically illustrates an embodiment of a C-shaped seal **256** located an upper portion of lower race **236** and near an upper portion of the rolling element **234** and an extending flange **282** of the trunnion **202** (shown in FIG. 7). The cross section of the C-shaped seal is seen to have its open end on an axial side of the seal, but it will be appreciated that other C-shaped seals in other locations on the assemblies described herein can have the open end of the c-shape on a radially outer end of the seal.

Turning now to FIGS. 9-27, an assembly process is shown illustrating various steps useful in constructing the trunnion-to-disk connection **224** (FIGS. 4, 6, and 7) of the integral trunnion and blade spar **200**. As will be appreciated, the steps outlined in the FIGS. 9-27 can be performed out of order owing to the separate nature of some steps and/or the involvement of different structures in the different steps. FIGS. 9 and 10 illustrate the attachment of the insert **252** to the trunnion **202**, where the insert **252** is an insert assembly having inserts **252a** and **252b**. Each of the inserts **252a** and **252b** are halves of an annular insert assembly, each of which include a groove **253** formed on an outer surface and into which the O-ring **254** can be inserted. The O-ring **254** can be used for construction purposes only to hold the inserts **252a** and **252b** in place, but in some embodiments can also serve a dual purpose of sealing as well.

FIGS. 11 and 12 illustrate the attachment of the lower race **236** (in which the downturned leg **261** is visible), the spanner nut **248** (FIGS. 4, 6, and 7) located about the upturned leg of the lower race **236**, C-shaped seal **256**, and an assembly scaffold **284** useful in closing out the construction for final installation of the gas turbine engine **100** (FIG. 1) on an aircraft. Various orders of assembly are contemplated of the components depicted in FIG. 11. One particular order is insertion of the assembly scaffold **284** onto the integral trunnion and blade spar **200**, followed by installation of the upper race **232** onto the integral trunnion and blade spar **200**, followed by installation of the rolling elements **234** (FIGS. 4, 6, and 7) and lower race **236** onto the

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integral trunnion and blade spar **200**. Any or all of the various component installations can be by press fit.

FIGS. 13-16 illustrate preparation of the fan disk **208** prior to receipt of the integral trunnion and blade spar **200** having been assembled with the components depicted in FIGS. 9-12. FIGS. 13 and 14 illustrate installation of the insert **250** into the trunnion aperture **258**. A hydraulic insert **286** used with the hydraulic lift **230** (FIGS. 4, 6, and 7) is also installed into an inner side wall of the trunnion aperture **258**. Both of the hydraulic insert **286** and the insert **250** can be installed using a press fit, but other techniques are also contemplated to fix these components in place in preparation for receipt of the integral trunnion and blade spar **200** having been assembled with the components depicted in FIGS. 9-12. FIGS. 15-16 illustrate the installation of the upper race **238** of the bottom bearing **228** (FIGS. 4, 6, and 7) into the trunnion aperture **258**. The upper race can be press fit into place to bear against a surface of the fan disk **208**.

FIG. 17 illustrates a pre-installation position where the integral trunnion and blade spar **200** having been assembled with the components depicted in FIGS. 9-12 is oriented such that the trunnion **202** can be inserted along the central aperture axis **260** of the trunnion aperture **258**. When the integral trunnion and blade spar **200** is inserted, it can be guided to a position illustrated in FIG. 18 such that the second end **243** of the trunnion **202** extends past an inner surface of the fan disk **208**. The position illustrated in FIG. 18 forms an access space **288** which permits installation of the rolling elements **240** (FIGS. 4, 6, and 7) and the lower race **242** (FIGS. 4, 6, and 7) for the bottom bearing **228** (FIGS. 4, 6, and 7). It will be appreciated that the access space **288** is defined by the foot **241** and inner surface of the fan disk **208**, such as overhang **239** and/or the upper race **238** of the bottom bearing **228**.

FIGS. 19-23 illustrate the installation of the rolling elements **240** into the access space **288**, installation of the lower race (in the form of lower races **242a** and **242b**), and subsequent movement of the integral trunnion and blade spar **200** (FIG. 17) into a final position via the hydraulic lift **230** (FIGS. 4, 6, and 7) and subsequent tightening of the spanner nut **248** (FIGS. 4, 6, and 7). FIG. 19 depicts several rolling elements **240** prior to insertion into the access space **288**. Given the large open interior **264** of the fan disk **208** into which the rolling elements **240** can be initially inserted, the rolling elements **240** are inserted axially (relative to the central aperture axis **260**) into a final resting position against the upper race **238** of the bottom bearing **228**. It will be appreciated that the foot **241** may require a radial movement (relative to the central aperture axis **260**) when in proximity to the foot **241**, but the final installation movement is substantially axial in direction. In some embodiments, a cage may be required to space and align the rolling elements correctly. Grease or other lubricant can be applied at this stage. After installation of the rolling elements **240**, the lower race **242** can be installed as illustrated in FIG. 21. In the illustrated embodiment the lower race **242** includes lower races **242a** and **242b**. The separate lower races **242a** and **242b** can be two halves of a complete lower race **242**, but other configurations of a split race assembly are contemplated. Once the lower races **242a** and **242b** are installed as shown in FIG. 22, the integral trunnion and blade spar **200** is ready for lowering into a final position via the hydraulic lift **230**. The hydraulic lift **230** as discussed above is capable of delivering hydraulic fluid into the cavity **246** (FIGS. 4, 6, and 7) and thereafter drive the integral trunnion and blade spar **200** along the central aperture axis **260**. It is contemplated that hydraulic pressure of about 2000 pounds per

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square inch (psi) can be used to achieve a full preload for the top bearing **226** (FIGS. **4**, **6**, and **7**) and bottom bearing **228** (FIGS. **4**, **6**, and **7**). Once a pressure target is achieved, the spanner nut **248** (FIGS. **4**, **6**, and **7**) can be hand tightened and, if desired, a final seating torque applied with a tool, which may or may not be a calibrated torque tool. When pressure is released from the hydraulic lift **230**, the spanner nut **248** holds the preload. FIG. **23** depicts a final installation position of the integral trunnion and blade spar **200**. Note the gap between the lower races **242a** and **242b** with respect to the fan disk **208** as illustrated in FIG. **22** (ready for final position) and as illustrated in FIG. **23** (final position).

FIGS. **24** and **25** illustrate a final close out device in the form of a grease cap **290** installed about the bottom bearing **228** (FIGS. **4**, **6**, and **7**) to enclose the bottom bearing **228** and discourage leakage of grease away from the bottom bearing **228** and also prevent the intrusion of debris into the bottom bearing **228**. The grease cap **290** can be affixed into place by using any variety of techniques including press fit around the second end **243** of the trunnion **202**.

As will be appreciated from the discussion herein, each trunnion-to-disk connection **224** (FIGS. **4**, **6**, and **7**) for each integral trunnion and blade spar **200** (FIGS. **4**, **6**, and **7**) is independently sealed such that each integral trunnion and blade spar **200** is in lubricant isolation from one another integral trunnion and blade spar **200** and the lubricant is not shared during operation of the gas turbine engine **100** (FIG. **1**) between any of the trunnion-to-disk connections **224**.

FIG. **26** discloses a method **292** of attaching an integral trunnion and blade spar **200** to a fan disk **208** of a gas turbine engine **100**. Step **294** includes inserting a trunnion **202** of the integral trunnion and blade spar **200** through a trunnion aperture **258** of a fan disk **208**, the trunnion **202** having a first trunnion end **237** with a foot **241** and a second trunnion end **243** opposite the first trunnion end **237**, the trunnion aperture **258** having a central aperture axis **260** transverse to a longitudinal axis **112** about the integral trunnion and blade spar **200**, and wherein the integral trunnion and blade spar **200** are rotated during operation of the gas turbine engine **100**. Step **296** includes locating the integral trunnion and blade spar **200** relative to the trunnion aperture **258** at a first position to define an access space **288** between the fan disk **208** and the integral trunnion and blade spar **200**. Step **298** includes installing a bottom bearing **228** into the access space **288** by moving the bottom bearing **228** along the central aperture axis **260** through an open interior **264** of the fan disk **208** and into the access space **288**.

The connection arrangement between the integral trunnion and blade spar **200** and the fan disk **208** of a gas turbine engine **100** provides various technical effects, including the ability to independently pack and lubricate the bottom bearing **228**. The higher radius ratio provided by the connection arrangement permits the large open interior **264** of the fan disk **208** into which the rolling elements **240** can be initially inserted into a final resting position against the upper race **238** of the bottom bearing **228**. Such a large open interior **264** can permit visual inspection of the rolling elements **240** as opposed to a blind insertion as in the prior art above.

Further aspects are provided by the subject matter of the following clauses:

A trunnion-to-disk connection for an open fan engine, the trunnion-to-disk connection comprising: a fan disk having an annular shape defined about a longitudinal axis, the fan disk defining a plurality of trunnion apertures and an inner surface, the inner surface defining a minimum radius from the longitudinal axis at a location aligned with the plurality

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of trunnion apertures, the plurality of trunnion apertures including a first aperture; a plurality of fan blades including a first fan blade having a central blade axis, the first fan blade further including an integral trunnion and blade spar positioned at least partially in the first aperture of the fan disk, a trunnion of the integral trunnion and blade spar including a first trunnion end including a foot and a second trunnion end opposite the first trunnion end, the foot having a bearing contact surface; and a bottom bearing captured between the bearing contact surface of the foot of the trunnion and the inner surface of the fan disk, the bearing contact surface defining a radius from the longitudinal axis less than the minimum radius of the inner surface of the fan disk; and wherein an access space is defined by the inner surface of the fan disk and the foot of the integral trunnion and blade spar, the access space configured to facilitate loading of the bottom bearing.

The trunnion-to-disk connection of the preceding clause, wherein the bottom bearing includes a rolling element captured between an upper race and lower race, wherein the lower race is a split race.

The trunnion-to-disk connection of any preceding clause, wherein each of the plurality of trunnion apertures includes a central aperture axis oriented coincident with the central blade axis.

The trunnion-to-disk connection of any preceding clause, wherein each of the plurality of trunnion apertures includes a central aperture axis, and wherein the integral trunnion and blade spar is structured to rotate about the central aperture axis to provide variable pitch to each of the plurality of fan blades.

The trunnion-to-disk connection of any preceding clause, wherein the foot extends transverse to the central aperture axis and away from the trunnion.

The trunnion-to-disk connection of any preceding clause, wherein the fan disk includes an inner abutment surface for bearing a load of at least one of the plurality of fan blades, which further includes a spanner nut located at the second trunnion end of the integral trunnion and blade, and wherein tightening of the spanner nut urges the foot to capture the bottom bearing between the foot and the inner abutment surface.

The trunnion-to-disk connection of any preceding clause, further comprising a top bearing and a foam insert, the foam insert positioned between the bottom bearing and the top bearing.

The trunnion-to-disk connection of any preceding clause, further comprising a top bearing, wherein a cavity is defined between the fan disk and the trunnion of the integral trunnion and blade, the cavity also defined between the top bearing and the bottom bearing.

The trunnion-to-disk connection of any preceding clause, wherein a cavity is defined between the fan disk and the trunnion of the integral trunnion and blade, and wherein the cavity is in lubricant isolation from another cavity defined between another integral trunnion and blade spar and the fan disk.

The trunnion-to-disk connection of any preceding clause, wherein each of the plurality of trunnion apertures includes a central aperture axis oriented to extend radially toward the longitudinal axis of the fan disk, and wherein the trunnion-to-disk connection further includes a grease port in fluid communication with the cavity, wherein the grease port is structured to deliver pressurized grease to the cavity to aid in forcing movement of the integral trunnion and blade spar along the central aperture axis.

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The trunnion-to-disk connection of any preceding clause, wherein the fan disk is coupled to a low pressure shaft of a gas turbine engine, and which further includes the gas turbine engine.

The trunnion-to-disk connection of any preceding clause, wherein one or both of the top bearing and the bottom bearing have rolling elements of a spherical type.

The trunnion-to-disk connection of any preceding clause, wherein one or both of the top bearing and the bottom bearing have rolling elements of a ball type.

The trunnion-to-disk connection of any preceding clause, wherein one or both of the top bearing and the bottom bearing have rolling elements of a cylindrical type.

The trunnion-to-disk connection of any preceding clause, wherein one or both of the top bearing and the bottom bearing have rolling elements of a tapered roller type.

The trunnion-to-disk connection of any preceding clause, wherein one or both of the top bearing and the bottom bearing have rolling elements of a needle roller type.

The trunnion-to-disk connection of any preceding clause, wherein a lower race of the top bearing includes at least one of an upturned leg, a base, and a downturned leg.

The trunnion-to-disk connection of any preceding clause, wherein a lower race of the top bearing includes a downturned leg, and which further includes an additional element adjacent the downturned leg.

The trunnion-to-disk connection of any preceding clause, which further includes at least one of an O-ring and a C-ring to discourage the passage of lubricant therethrough.

A powerplant system comprising: gas turbine engine, a fan section having a fan blade with an integral trunnion and blade spar, a fan disk for coupling with the fan blade, and a trunnion-to-disk connection used to connect the fan disk with the fan blade.

A method of attaching an integral trunnion and blade spar to a fan disk of a gas turbine engine, the method comprising: (a) inserting a trunnion of the integral trunnion and blade spar through a trunnion aperture of the fan disk, the trunnion having a first end with a foot and a second trunnion end opposite the first trunnion end, the trunnion aperture having a central aperture axis transverse to a longitudinal axis about which the integral trunnion and blade spar are rotated during operation of the gas turbine engine; (b) locating the integral trunnion and blade spar relative to the trunnion aperture at a first position to define an access space between the fan disk and the integral trunnion and blade spar; and (c) installing a bottom bearing into the access space by moving the bottom bearing along the central aperture axis through an open interior of the fan disk and into the access space.

The method of the preceding clause, wherein the bottom bearing is separable and having an upper race, a lower race, and a plurality of rolling elements, wherein the installing the bottom bearing includes inserting the upper race into the access space independent of installation of the lower race, inserting each rolling element of the plurality of rolling elements into the access space, and inserting the lower race into the access space.

The method of any preceding clause, wherein the inserting the upper race includes connecting the upper race with an overhang formed in the fan disk.

The method of any preceding clause, wherein a cavity is defined between the fan disk and the trunnion of the integral trunnion and blade, a top bearing and the bottom bearing, the method further comprising pressurizing the cavity to urge the foot into compression with the lower race of the bottom bearing.

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The method of any preceding clause, further comprising tightening a spanner nut to the top bearing at the second trunnion end to compress the bottom bearing between the foot of the trunnion and an inner abutment surface of the fan disk, and wherein the pressurizing includes injecting the cavity with a lubricant, and wherein the pressurizing occurs before the tightening of the spanner nut.

The method of any preceding clause, further comprising installing a foam insert into the cavity prior to the pressurizing.

The method of any preceding clause, further comprising translating the top bearing along the central aperture axis as a result of the pressurizing.

The method of any preceding clause, further comprising obtaining another integral trunnion and blade spar and repeating steps (a) through (d) for the another integral trunnion and blade spar.

The method of any preceding clause, further comprising enclosing the bottom bearing associated with the integral trunnion and blade spar and enclosing the bottom bearing associated with the another integral trunnion and blade spar such that the bottom bearing of each of the integral trunnion and blade spar and the another integral trunnion and blade spar are in lubricant isolation from one another.

This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

We claim:

1. A method of attaching an integral trunnion and blade spar to a fan disk of a gas turbine engine, the method comprising:

- (a) inserting a trunnion of the integral trunnion and blade spar through a trunnion aperture of the fan disk, the trunnion having a first end with a foot and a second trunnion end opposite the first trunnion end, the trunnion aperture having a central aperture axis transverse to a longitudinal axis about which the integral trunnion and blade spar are rotated during operation of the gas turbine engine;
- (b) locating the integral trunnion and blade spar relative to the trunnion aperture at a first position to define an access space between the fan disk and the integral trunnion and blade spar; and
- (c) installing a bottom bearing into the access space by moving the bottom bearing along the central aperture axis through an open interior of the fan disk and into the access space.

2. The method of claim 1, wherein the bottom bearing is separable and having an upper race, a lower race, and a plurality of rolling elements, wherein the installing the bottom bearing includes inserting the upper race into the access space independent of installation of the lower race, inserting each rolling element of the plurality of rolling elements into the access space, and inserting the lower race into the access space.

3. The method of claim 2, wherein the inserting the upper race includes connecting the upper race with an overhang formed in the fan disk.

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4. The method of claim 2, wherein a cavity is defined between the fan disk and the trunnion of the integral trunnion and blade, a top bearing and the bottom bearing, the method further comprising pressurizing the cavity to urge the foot into compression with the lower race of the bottom bearing.

5. The method of claim 4, further comprising tightening a spanner nut to the top bearing at the second trunnion end to compress the bottom bearing between the foot of the trunnion and an inner abutment surface of the fan disk, and wherein the pressurizing includes injecting the cavity with a lubricant, and wherein the pressurizing occurs before the tightening of the spanner nut.

6. The method of claim 4, further comprising installing a foam insert into the cavity prior to the pressurizing.

7. The method of claim 4, further comprising translating the top bearing along the central aperture axis as a result of the pressurizing.

8. The method of claim 1, further comprising obtaining another integral trunnion and blade spar and repeating steps (a) through (d) for the another integral trunnion and blade spar.

9. The method of claim 8, further comprising enclosing the bottom bearing associated with the integral trunnion and blade spar and enclosing the bottom bearing associated with the another integral trunnion and blade spar such that the bottom bearing of each of the integral trunnion and blade spar and the another integral trunnion and blade spar are in lubricant isolation from one another.

10. The method of claim 1, wherein the bottom bearing includes a rolling element captured between an upper race and a lower race, wherein the lower race is a split race.

11. The method of claim 1, wherein each of the plurality of trunnion apertures includes a central aperture axis oriented coincident with the central blade axis.

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12. The method of claim 1, wherein each of the plurality of trunnion apertures includes a central aperture axis, and wherein the integral trunnion and blade spar is structured to rotate about the central aperture axis to provide variable pitch to each of the plurality of fan blades.

13. The method of claim 1, wherein the gas turbine engine further comprises a top bearing and a foam insert, the foam insert positioned between the bottom bearing and the top bearing.

14. The method of claim 1, wherein the gas turbine engine further comprises a top bearing, wherein the cavity is defined between the fan disk and the trunnion of the integral trunnion and blade, the cavity also defined between the top bearing and the bottom bearing.

15. The method of claim 14, wherein the cavity is defined between the fan disk and the trunnion of the integral trunnion and blade, and wherein the cavity is in lubricant isolation from another cavity defined between another integral trunnion and blade spar and the fan disk.

16. The method of claim 14, wherein each of the plurality of trunnion apertures includes the central aperture axis oriented to extend radially toward the longitudinal axis of the fan disk, and wherein the trunnion-to-disk connection further includes a grease port in fluid communication with the cavity, wherein the grease port is structured to deliver pressurized grease to the cavity to aid in forcing movement of the integral trunnion and blade spar along the central aperture axis.

17. The method of claim 1, wherein the fan disk is coupled to a low pressure shaft of a gas turbine engine, and which further includes the gas turbine engine.

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